Victorian Government Department of Energy, Environment and Climate Action

Water supply and demand in the Southern Murray-Darling Basin

An assessment of future water availability and permanent horticulture irrigation water demand

31 March 2026

Document distribution

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Prepared for

Victorian Government Department of Energy, Environment and Climate Action

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About this report

This report presents key findings from an analysis of water availability and demand in the southern Murray-Darling Basin. It is an update to a 2022 report and examines recent water usage patterns across the connected Murray and lower Murray regions, along with potential future use under different levels of permanent horticulture development. It also compares water demand from permanent horticulture with the volume of allocations available for consumptive use under five different water availability scenarios. WSP relies on data obtained from multiple third-party sources. Consequently, any information in respect to water availability and demand shall be subject to the accuracy and limitations of the data obtained from third-party sources. The appendices present full details of the data, data sources and assumptions used in the analysis presented in this report.

Citation

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Abbreviations and glossary

Abbreviations

Abbreviation	Description
ABA	Almond Board of Australia
AG DCCEEW	Australian Government Department of Climate Change, Energy, Environment and Water
DEECA	Victorian Department of Energy, Environment and Climate Action
DEW	South Australian Department for Environment and Water
EPBC Act	Environment Protection and Biodiversity Conservation Act
NSW DCCEEW	New South Wales Department of Climate Change, Energy, Environment and Water
GL	Gigalitre
GS	General Security
HRWS	High Reliability Water Share
HS	High Security
IVT	Intervalley trade
LRWS	Low Reliability Water Share
LTDLE	Long-term diversion limit equivalent
MDB	Murray-Darling Basin
MDBA	Murray-Darling Basin Authority
ML	Megalitre
NSW	New South Wales
SA	South Australia
VIC	Victoria

Glossary of terms

Term	Explanation
Entitlement on issue	The total number of water entitlements for an entitlement reliability and water course.
Carryover	Unused water that can be retained in water storages from one year to the next by water owners.
Horticultural headroom	The difference between water available to all irrigated agriculture and water demand from permanent horticulture.

Term	Explanation
Long-term diversion limit equivalent factor	Also known as Cap factors, these were established to accurately assess how much water has been recovered for the environment, and to guide future water recovery decisions. It offers a consistent accounting system for tracking water recovery against targets.
Permanent horticulture	Inclusive of perennial fruit and nut trees (such as almonds, stone fruit, pome fruit and citrus), grapevines (both table grapes and wine grapes) and berry fruits.
Water allocation	The volumes of water allocated to water entitlement holders during the water year (1 July to 30 June). They are a physical good analogous to a commodity and are extracted from water courses and applied as inputs to production or the environment.
Water allocated that is available for consumptive use	Refers to the total volume of water allocated by state governments to water entitlement holders for consumptive use e.g., irrigation during the water year (1 July to 30 June). This excludes environmental holdings (Federal and state) and urban water holdings. It includes entitlement purchases by the Australian government under the Aboriginal Water Entitlement Program. Measured as an aggregate volume at the end of the water year (30 June).
Water entitlement	Water entitlements are ongoing rights to receive an annual share of available water resources in a consumptive pool as established in a specific river system, catchment, or aquifer. Entitlements are generally secure, tradeable, divisible and mortgageable in the same way as land.

Executive summary

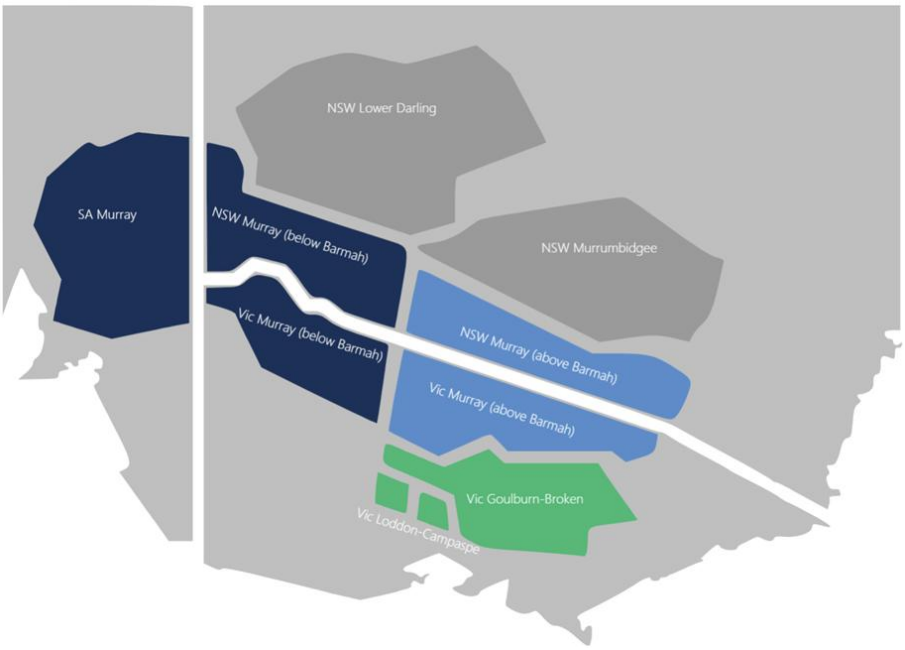
This report provides an update to earlier assessments of water supply availability and permanent horticultural demand in the southern Murray-Darling Basin (Version 1, Aither 2019; Version 2, Aither 2020; Version 3, Aither 2022). The analysis in this report has been updated in collaboration with state government water agencies in Victoria (Vic), New South Wales (NSW), and South Australia (SA), as well as the Commonwealth Environmental Water Holder (CEWH).

The report contributes to efforts to enhance water market transparency and support participants in making informed decisions by assessing the pressure on water available for irrigation in the southern Murray-Darling Basin and reflecting changes since the report was produced. Since the 2022 report, irrigation development in the region has continued to progress, and more recently, improved data has become available. During this period, there have also been changes to water markets in the southern Murray-Darling Basin, including the Commonwealth's reintroduction of voluntary water purchases to meet Basin Plan water recovery targets.

The connected Murray and lower Murray regions

The **connected Murray** region is defined as all major water systems within the southern Murray-Darling Basin with connectivity to the Murray River, **excluding NSW Murrumbidgee and Lower Darling**. The specific systems captured within the connected Murray region are shown by all coloured systems in Figure 1. The systems coloured dark blue signify the **lower Murray** region. Accessing water across the connected Murray is subject to trade constraints – most notably trading water into the lower Murray from other valleys.

Figure 1 Water systems in the southern Murray-Darling Basin showing the connected Murray (all coloured systems) and lower Murray sub-regions (dark blue).



Key Findings

Below is a summary of the results and key messages from the 2025 analysis. Overall, the results of this report indicate a slight improvement in water availability compared to the 2022 analysis. However, the results remain similar, and key messages remain consistent: **market participants need to be prepared for the potential implications of years with limited water availability.**

Water demand for permanent horticulture continues to increase

As existing plantings continue maturing, and new developments progress, there has been an increase in the baseline water demand from permanent horticulture since 2020-21 (Table 1). It is, however, noted that water use varies from year to year, and 2024-25 had warmer and drier conditions in the Murray than 2020-21.

Table 1 Water demand estimate (baseline existing permanent plantings)

	Connected Murray			Lower Murray		
	(2020-21)	(2024-25)	Difference	(2020-21)	(2024-25)	Difference
Water demand estimate	1,128 GL	1,239 GL	+111 GL	1,049 GL	1,132 GL	+ 83 GL
Breakdown by state						
New South Wales	*180 GL*	*159 GL*	*- 21 GL*	*167 GL*	*138 GL*	*- 29 GL*
Victoria	*589 GL*	*633 GL*	*+ 44 GL*	*523 GL*	*546 GL*	*+ 23 GL*
South Australia	*359 GL*	*448 GL*	*+89 GL*	*359 GL*	*448 GL*	*+ 89 GL*

Supply has also increased since the last report

The amount of water available has increased marginally since the last report. This is because of the distribution of the irrigators' share of water recovered through the Connections Project[1], which the Victorian Government committed to providing irrigators in the Goulburn-Murray Irrigation District (GMID). It must be noted that water availability does not include Australian government water recovery associated with the Voluntary Water Purchase Program for the 450 GL, which will reduce future irrigation water availability. This is examined separately further below.

Table 2 illustrates the effect on the estimated water supply in the lower Murray under dry to extreme dry scenarios. Although supply has improved relative to demand, the net effect is marginal.

[1] https://yoursay.gmwater.com.au/connections-project

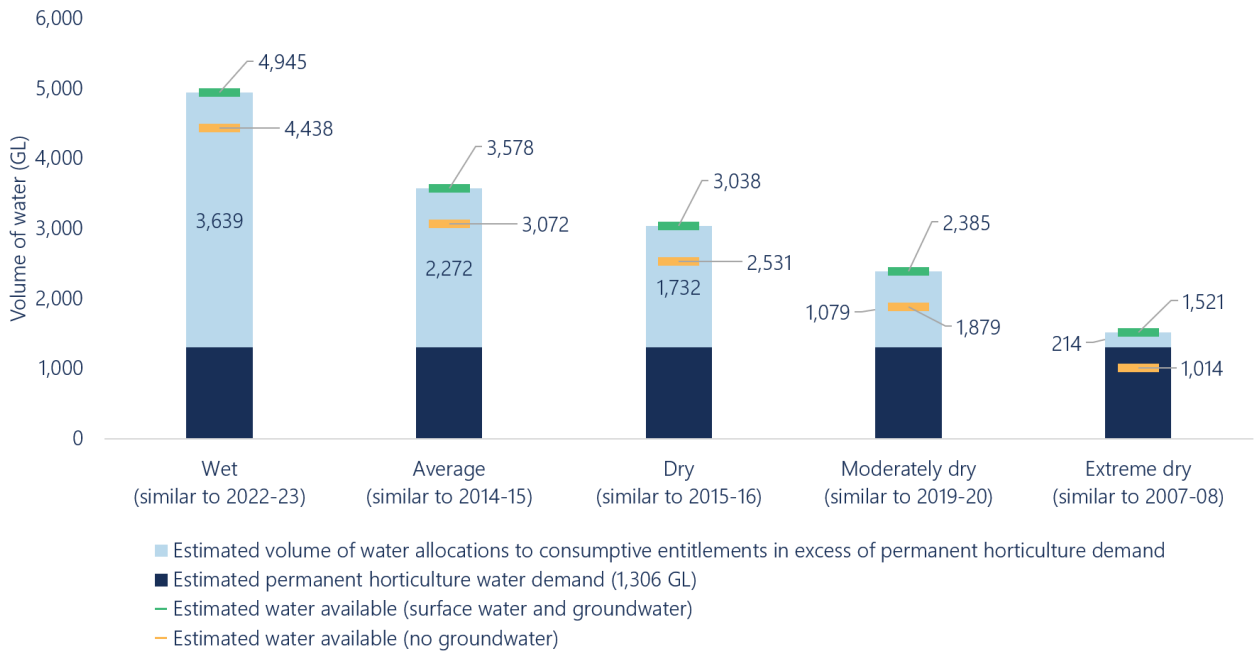
Table 2 Estimated dry to extreme dry water supply scenarios

	Connected Murray			Lower Murray		
	(2020-21)	(2024-25)	Difference	(2020-21)	(2024-25)	Difference
Dry (similar to 2015-16)	3,007 GL	3,038 GL	+ 31 GL	1,364 GL	1,372 GL	+ 8 GL
Moderately Dry (similar to 2019-20)	2,359 GL	2,385 GL	+ 26 GL	1,067 GL	1,072 GL	+ 5 GL
Extreme dry (similar to 2007-08)	1,502 GL	1,521 GL	+ 19 GL	473 GL	477 GL	+ 4 GL

The risk remains that permanent horticultural demand in the lower Murray will exceed supply under extreme dry conditions.

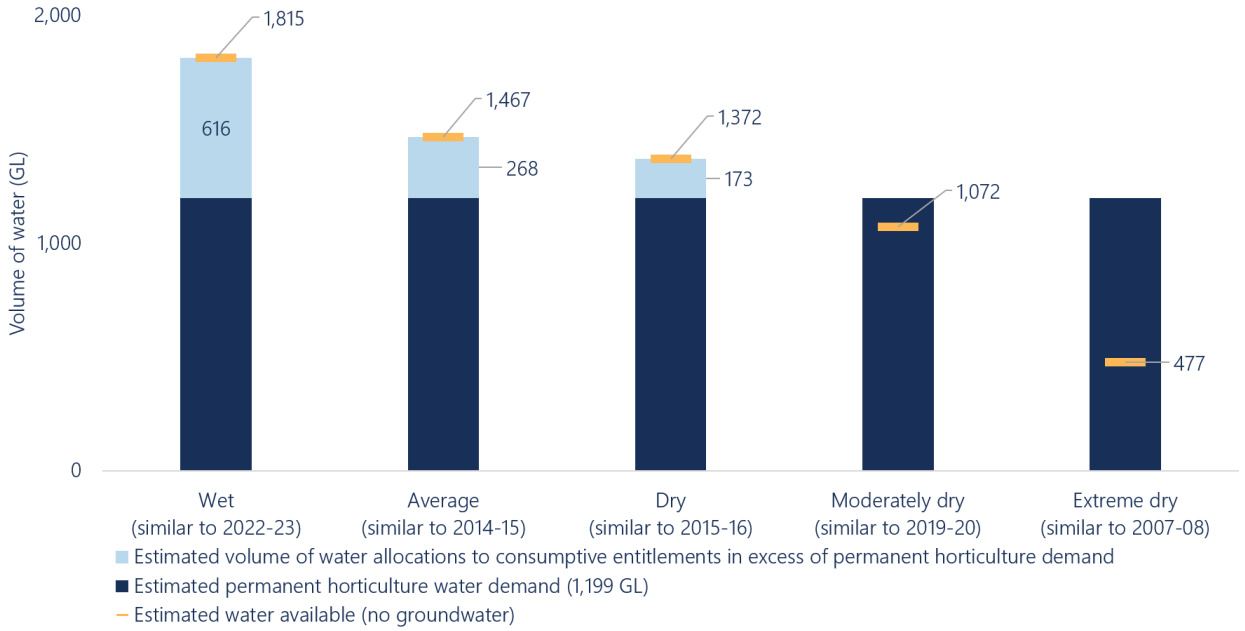
In the connected Murray, water supply (including groundwater) exceeds demand from existing permanent horticulture (at full maturity) under all climatic scenarios (Figure 2). However, accessing water across the connected Murray is subject to trade constraints, these restrictions have not been considered in modelling water supply in the connected Murray.

Figure 2 Water availability scenarios and existing permanent horticulture water demand (permanent plantings at full maturity), connected Murray region



Consistent with previous analysis, in the lower Murray, supply is exceeded by existing permanent demand (at full maturity) under both extreme dry (-722 GL) and moderately dry conditions (-127 GL) (Figure 3).

Figure 3 Water availability scenarios and existing permanent horticulture water demand (permanent plantings at full maturity), lower Murray region



This highlights the importance of this analysis for water market participants in the lower Murray. A repeat of conditions similar to the Millennium Drought would be extremely challenging for permanent horticulturalists in the lower Murray. This highlights the continuing importance of intervalley trade and carryover as a risk-mitigation tool for water users in the lower Murray.

The expansion of permanent horticulture within the lower Murray continues; however, new development has slowed markedly.

Although the overarching findings of this report align with those of the previous report, the analysis shows that the 5-year projections for 2022[2] have not yet materialised to the extent projected over the last ~5 years. This is illustrated in Table 3 and Figure 4, where the actual increase to date (ha) is just over 25 per cent of the total 5-year projected increase in the 2022 report.

A key difference between the 2025 and 2022 analysis on projected plantings was the 2022 analysis used two scenarios to distinguish between small/medium and large plantings. However, in 2025, general market sentiment drove a reduction in planned plantings, particularly in wine grapes – changing the approach in this report.

[2] The 2022 report analysis used two scenarios: Scenario A included all planned plantings with a high likelihood of proceeding under 1,000 hectares, and Scenario B included all planned plantings. In this report, WSP compares Scenario B from the 2022 analysis with the 2025 analysis

Table 3 Projected 5-year change in future plantings in the lower Murray (ha) compared to the 5-year projection in the 2022 report (and what actually materialised in the ~5 years between the 2022 report and 2025 report (this report)).

Lower Murray	Projected 5-year change in plantings from the 2022 report	Actual ~5-year change in plantings from the 2022 report	Projected 5-year change in plantings from the 2025 report
Almonds (ha)	+16,039[3]	3,730	+3,452
Citrus (ha)	+5,764	2,265	+1,308
Grapes (ha)	+10,706	-205	-1,899
Total projected increase (ha)	**+31,852**	**5,790**	**+2,861**
Total projected increase in water demand (GL)	**+409**		**+45**

[3] Includes 657 hectares of "other nuts" from the 2022 analysis.

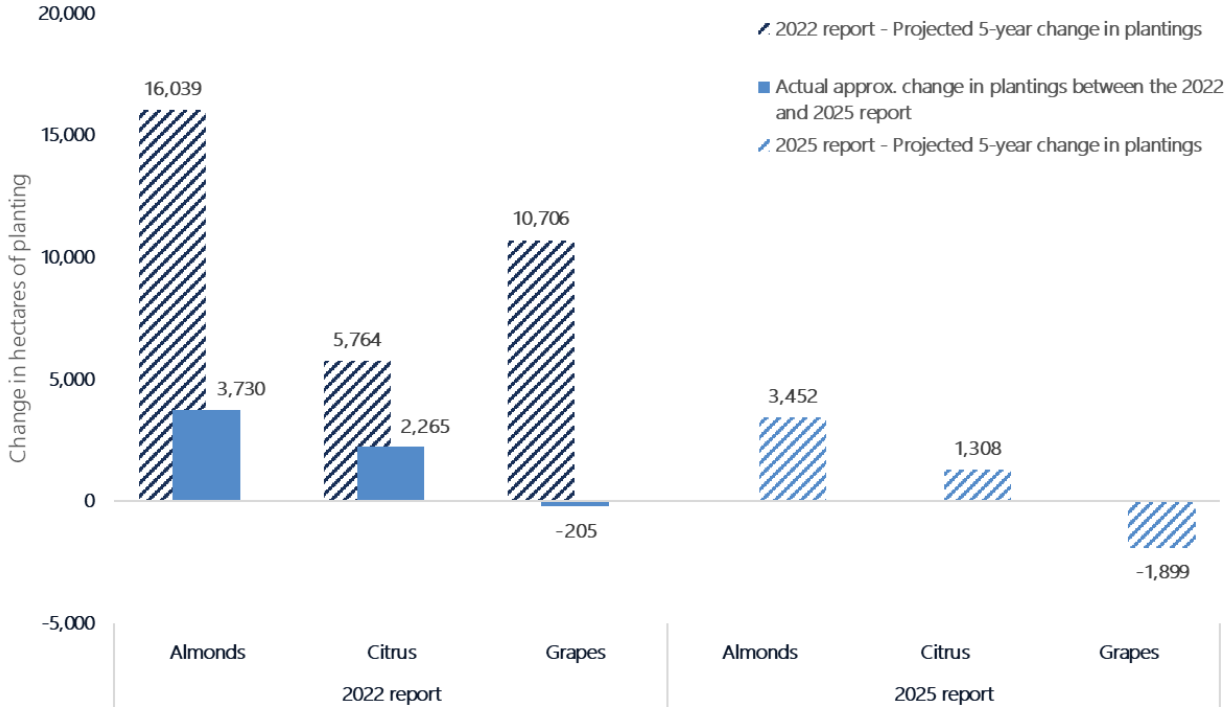


Figure 4 Projected change in future plantings (hectares) in the lower Murray in the next 5 years compared to what was projected in the 2022 report (and what actually materialised between the 2022 report and 2025 report).[4]

There are several likely reasons for the slower than expected growth. Firstly, the influence of broader economic factors, such as a three-year decline in almond prices, rising interest rates and challenging trade dynamics. Related to this is the sharp decline in demand for winegrapes, driven by past trade tensions with China and a global oversupply. This has reduced the area planted to winegrapes, which is somewhat masked in the grape category by a simultaneous rise in table grape area. Nevertheless, across the three states, the area planted to grapes has decreased by almost 2,000 ha since the last report. Based on industry information for this report, the area planted to grapes is expected to decrease further over the next five years, reflecting ongoing challenges in the winegrape industry.

At the time of writing the 2022 report, industry sources were not predicting a significant change to winegrape plantings. Supporting this was Wine Australia's 2022 South Australian Winegrape Crush Survey[5], which reported a 0.03% net decrease in vineyard area, noting that SA represents approximately 52% of total winegrape area in Australia.

Secondly, government policy changes, such as the Commonwealth Government resuming water recovery to deliver Basin Plan objectives, create uncertainty over future water availability. Also, starting in 2019, the Victorian Government ceased issuing any further 'extraction share' (the daily rate that private river diverters can take when there is not enough water to meet all demands) in the lower Murray. This was formalised as a cap on total extraction share with the introduction of the

[4] 2022 projected almonds figure includes 657 hectares of "other nuts" from the 2022 analysis.
[5] https://vinehealth.com.au/wp-content/uploads/SA-full-report-2022.pdf

"Place of Take" legislative framework in late 2023. This has likely affected development in Victoria, whilst development has continued in NSW and South Australia, as shown in Table 4.

Table 4 Lower Murray existing area planted at full maturity

Lower Murray	2022 Report	2025 Report (% change from 2022 report)
NSW		
Almonds (ha)	5,125	5,855 (+14%)
Citrus (ha)	2,940	4,020 (+37%)
Grapes (ha)	9,095	9,240 (+2%)
VIC		
Almonds (ha)	26,405	26,365 (-0.2%)
Citrus (ha)	4,715	5,645 (+20%)
Grapes (ha)	21,550	20,955 (-3%)
SA		
Almonds (ha)	11,460	14,480 (+26%)
Citrus (ha)	6,455	6,740 (+4%)
Grapes (ha)	28,525	26,590 (-7%)
Total Area Planted (ha)	**114,100**	**119,890 (+5%)**
Total Water Demand – existing plantings at full maturity (GL)	**1,196**	**1,199 (+0.25%)[6]**

Additional known permanent plantings planned above the Barmah Narrows and in the Goulburn are unlikely to have a material impact on the available headroom in the connected Murray.

Overall, the projected demand estimates discussed earlier in Table 3 indicate what might occur if existing plantings reach maturity, planned developments are completed, and there are no unforeseen planting / water demand reductions in other horticultural sectors. Nonetheless, there is considerable uncertainty about future outcomes, particularly regarding permanent horticultural

[6] Estimated increase in water demand is lower than the increase in planting area due to planting area data and water usage data being collated from different sources. For further detail refer to Appendix C and D.

developments at various planning stages. While there remains strong interest among developers of permanent horticultural assets in the lower Murray (below the Barmah Narrows), uncertainty remains about which projects will proceed. Decision-making for individual enterprises will depend on a complex mix of factors; for this reason, analyses such as those in this report must be regularly updated.

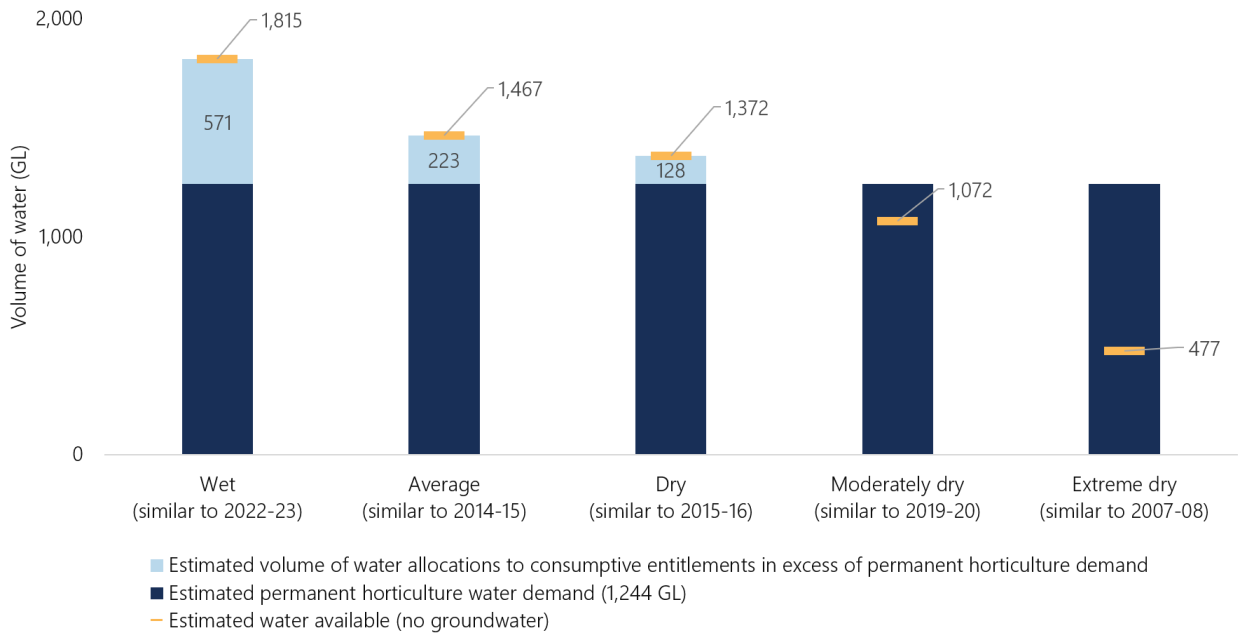
In the lower Murray, during future extreme dry years, water demand from existing permanent horticulture (at full maturity) may exceed the region's available water by two and a half times.

This report's analysis reinforces the previous finding that, during moderately dry and extreme dry years, there will likely be no horticultural headroom available in the lower Murray. While the analysis confirms that demand from permanent horticulture is increasing, the rate of growth has slowed compared with previous reports.

Considering the maturity profile of the existing permanent horticulture plantings, water demand by these crops at full maturity in the lower Murray region could reach 1,199 GL. This estimate is based on the age distribution of permanent horticulture crops and their respective water application rates over time. It shows a 3 GL increase compared with the 2022 estimate, which projected a significant rise in demand from planned developments across Victoria, New South Wales, and South Australia, ranging from 175 to 409 GL at full maturity. The 2025 analysis indicates smaller growth, with a 45 GL increase in water demand, noting that grapes are expected to decline by 19 GL (driven by wine grapes).

With planned plantings at full maturity included, in extreme dry and moderately dry years without carryover or intervalley trade into the lower Murray, there will likely be no horticultural headroom. During a moderately dry year, demand from permanent horticulture may exceed availability by around 16 per cent, with estimated demand of 1,244 GL and surface water availability of 1,072 GL.

Figure 5 Water availability scenarios and projected permanent horticulture water demand, including developments with a high chance of proceeding, lower Murray region

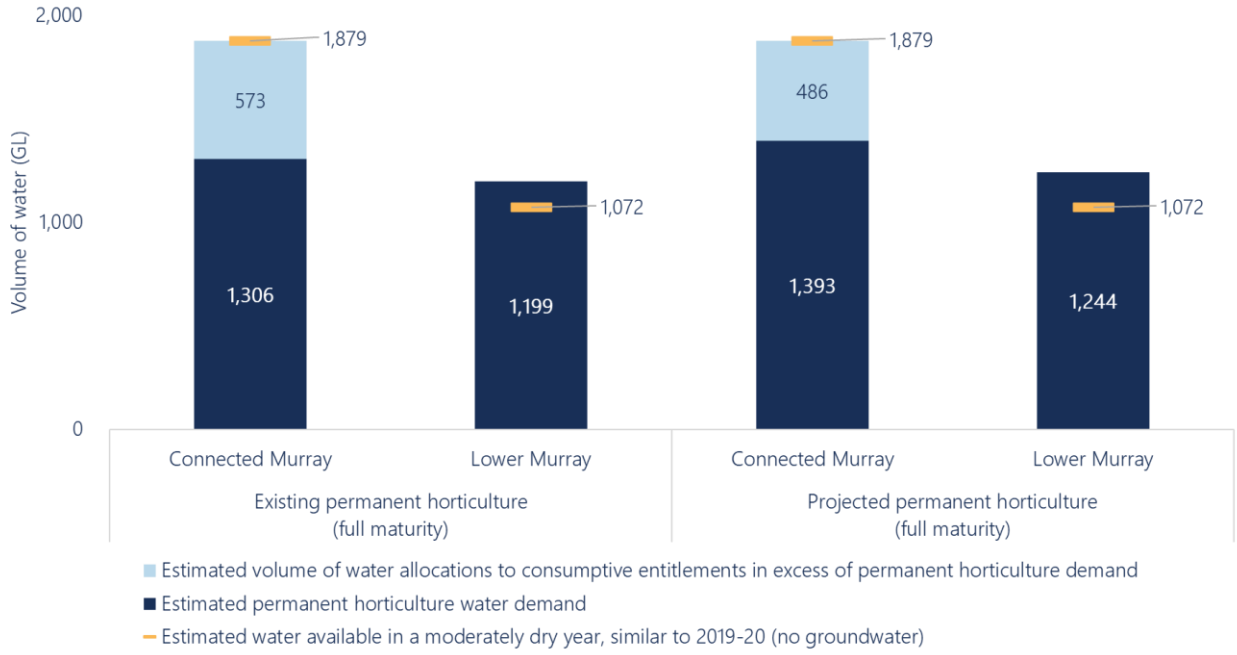


Inter-valley trade and carryover remain key risk-management tools for all irrigated industries, even under a moderately dry water-availability scenario.

It is important to acknowledge that intervalley trade and carryover are deliberately not considered directly in the modelling for this or past reports. The results continue to emphasise to market participants the crucial importance of access to the wider connected southern Murray-Darling Basin water market through inter-valley trade from the Goulburn, Murrumbidgee, and upper Murray. The findings also show that carryover can help reduce some, but not all, of this risk.

Figure 6 illustrates that, in a moderately dry year such as 2019-20, whilst there is a shortfall in the lower Murray, there may be sufficient water across the connected Murray. However, intervalley trade opportunities are likely to be highly contested, and a water strategy that relies on them carries some risk. Carryover from the previous water year can help mitigate this risk and can be combined with intervalley trade towards the end of the water year, when there is usually a greater opportunity to move water into the lower Murray.

Figure 6 Water availability and demand in the lower Murray, moderately dry water availability scenario compared to existing and future demands



Commonwealth water recovery could further heighten pressure on horticultural headroom in both the connected and lower Murray regions.

Commonwealth water recovery programs are currently targeting entitlements in the southern Murray-Darling Basin to contribute to the 450 GL/y of additional environmental water.[7] To model the impact of reduced supply from future water recovery on horticultural headroom, a **hypothetical** scenario was developed to assess consumptive water availability after the Commonwealth has met this target.

The volume required to be recovered to meet the difference between the consumptive entitlement on issue for the southern Murray-Darling Basin in 2024-25 and full recovery of the 450 GL/y target was estimated at 363 GL/y. This assumes that purchases are made in proportion to the Commonwealth Environmental Water Holder's existing water portfolio.[8]

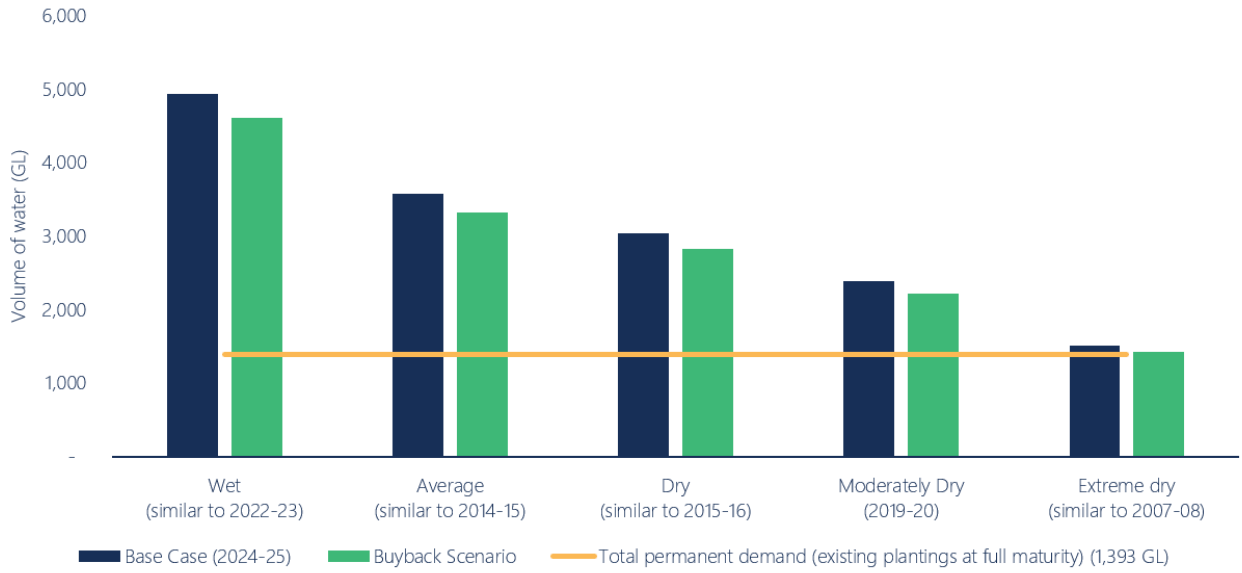
- The remaining 363 GL/y to be purchased corresponds to a 91 GL to 329 GL decrease in volume of water available in the connected Murray depending on the climate of any given year. Wetter years have greater volume reductions because more water is available. The most notable decrease is under the extreme dry scenario in the connected Murray (with groundwater included), where horticultural headroom decreases by 71% to 37 GL (Figure 7). When groundwater is excluded, this corresponds to a 7% to 9% decrease in water availability across scenarios.

[7] Department of Climate Change, Energy, the Environment and Water, 2025, Voluntary Water Purchase Program for the 450 GL (Restoring our Rivers), https://www.dcceew.gov.au/water/policy/water-recovery/government-water-purchasing/voluntary-restoring-our-rivers

[8] This is a hypothetical scenario that does not reflect the position of either the Commonwealth government, Basin State governments or WSP on what entitlements will be a target of future water recovery efforts. This has been developed for illustrative purposes only for readers of this report.

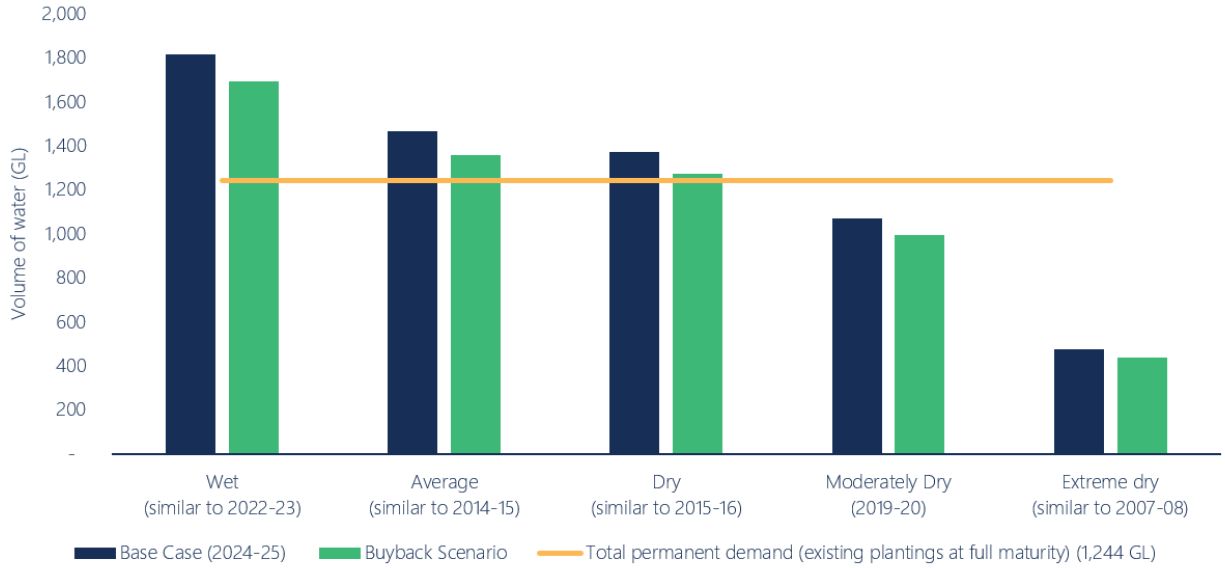
- In the lower Murray, the volume of water available is estimated to decrease by 37 GL to 124 GL depending on the climate in a given year. This corresponds to a 7% to 8% decrease in the lower Murray (excluding groundwater), with drier years having the largest percentage decreases.

Figure 7 Connected Murray, comparison of 2024-25 consumptive water availability with a buyback scenario against total existing demand at full maturity.



The impact on the lower Murray is significantly greater. The reductions push the moderately dry scenario into a significant supply shortfall; headroom declines by 45%, resulting in a 250 GL shortfall. Under a dry scenario, the headroom drops to just 27 GL (a 79% decrease) (Figure 8).

Figure 8 Lower Murray, comparison of 2024-25 consumptive water availability with a buyback scenario against total existing demand at full maturity.



1. Introduction

1.1 Scope

Ricardo, a member of WSP, was engaged by the Victorian Department of Energy, Environment and Climate Action (DEECA) to update the analysis reported in **Water supply and demand in the southern Murray-Darling Basin (2022 update): An assessment of future water availability and permanent horticulture irrigation water demand** (Aither 2022). This was done in collaboration with state and federal governments. The analytical approach to investigating the changing dynamics of water availability and demand under different inflow conditions remains consistent with previous reports (Aither 2019; 2020; 2022). The data sources and approach used in the 2025 report are most closely aligned with those used in the 2020 and 2022 reports (Aither 2020; 2022).

1.2 Purpose

Water market transparency remains a key focus of Murray–Darling Basin reforms and broader water policy. Following the **ACCC's Murray-Darling Basin water markets inquiry final report (2021)**, which underscored the need for accessible and reliable market information, Basin states undertook projects in 2021–22 to improve data quality and accessibility. Since then, notable progress has been made under the **Water Market Reform Roadmap** (Quinlivan, 2022), which outlined 23 recommendations for practical, cost-effective reforms. The **Water Amendment (Restoring Our Rivers) Act 2023** introduced new measures to improve the integrity and transparency of water markets, and a new **Water Markets Intermediaries Code** came into effect in July 2025.

Our analysis builds on the work from the 2019, 2020 and 2022 reports to support water resource management decisions across the southern Murray-Darling Basin. It provides current information on water availability and permanent horticultural use. It illustrates how the gap between the total volume of water available for consumptive use and the volume that could be used by permanent horticulture may change across five climatic scenarios and under different horticultural development paths.

This report presents the findings of the updated 2025 analysis. The results are presented for easy comparison with earlier reports. To support transparency and provide background for the analysis, the data, sources, and assumptions are clearly detailed in the appendices (Appendix B, Appendix C, Appendix D).

This analysis supports informed decision-making by water market participants and other interested stakeholders, including:

1. Entitlement holders and water users across all irrigated agricultural industries in the southern Murray-Darling Basin who are making decisions about water availability and pricing expectations.
2. New water market entrants and existing horticulturalists planning to expand their operations in the southern Murray-Darling Basin who need to assess water costs and availability.
3. Government decision-makers, committees, and policy makers responsible for water market reforms.

1.3 Rationale for updating the analysis

The 2019, 2020, and 2022 water availability and permanent horticulture demand reports contributed to government efforts to enhance water market transparency. The 2025 report continues to provide comparable information on potential future water availability scenarios and their alignment with water demand from permanent horticulture plantings in the southern Murray-Darling Basin. This report utilises water usage data from 2024-25, addresses the potential impact of the Australian Government re-commencing water purchase in 2025, and examines how changing trade rules may affect the water market.

1.3.1 New data to help refine the analysis

In this version of the analysis, the following new data were available and used:

- 2024-25 water usage data from the state governments, which provides a good opportunity to test the sensitivity of the results from the 2022 analysis.
    - The 2022 analysis uses 2020-21 as the baseline year, which was relatively cool, particularly during the summer irrigation period. This meant permanent horticulturalists were less reliant on irrigation water and likely to have lower water application rates.
    - The 2025 analysis uses 2024-25 water usage data as the baseline year. Growing conditions in 2024-25 were markedly different from those in 2020-21. 2024-25 was characterised by relatively dry and hot conditions, with low rainfall. As a result, compared with 2020-21, water application rates by permanent horticulturalists were likely higher in 2024-25.
    - The new data captures the maturation of horticultural plantings and recent declines in the wine grape sector.
- Updated crop area data to support estimates of water demand from existing permanent horticulture at full maturity, including the Edward-Wakool system and irrigated areas along the Murray River upstream to the Barmah Narrows (SunRISE Mapping and Research 2025). This would likely capture the continued maturation of existing horticultural plantings and recent declines in wine grape plantings due to ongoing economic and trade headwinds.

These data improvements helped refine the analysis and increased confidence in the estimates.

1.3.2 Commencement of water purchases by the Australian government to meet environmental targets

The *Restoring Our Rivers Act 2023* introduced legislative amendments to facilitate the full implementation of the Basin Plan. These changes extended the timeframes for delivering the Plan, including:

- The deadline to complete Sustainable Diversion Limit Adjustment Mechanism (SDLAM) supply projects has been extended to 31 December 2026.
- Additional time for new SDLAM projects to be proposed by June 2025.
- The deadline for the Basin Plan to be delivered, including recovery of the 450 GL/y additional water for the environment, is 31 December 2027 (previously 30 June 2024).

The amendments also repealed the 1,500 GL cap on Australian government water purchases, paving the way for additional purchases, which have since commenced, with planned future purchases continuing on a strategic timeline.

- In July 2024, the Australian government opened the first tender for water purchases towards the 450 GL/y target, seeking 70 GL across five catchments in the southern connected Basin. The November 2025 update marked the tender as closed and complete, with the Commonwealth finalising contractual processes, bringing the total to 51.2 GL/y under the 'Selected catchments' tender.
- In September 2024, the Australian government opened two Expressions of Interest, inviting parties interested in selling water rights in the 'Remaining southern connected Basin' (EOI1) and 'Large portfolios of water' (EOI2) to submit responses. In February 2025, the Australian government commenced purchases under the EOI2 program. The November 2025 announcement expanded the amount of water to be purchased towards the 450 GL target to 300 GL. Both programs remain ongoing as of December 2025.

The net result of the government's water purchases has been the redirecting of water from the consumptive pool to the environment, reducing the total volume of water available to irrigators and other water market participants.

1.3.3 Outline how the water market is evolving and how water demand is changing over time

The impact of new developments and the maturation of existing permanent horticulture plantings, alongside a decline in wine grapes, on market prices and water availability remains a topic of interest for water market participants and government authorities alike. As water markets in the southern Murray-Darling Basin continue to evolve, regular updates on supply and demand dynamics are needed to support decision-making. Such updates also reflect the Victorian Government's continued commitment to improving water market transparency.

Furthermore, new long-term trade and operating rules for the Goulburn to Murray Trade Rule came into effect on 1 July 2022 to balance the demands of water market participants and the environment. It has now been over three years since the rule came into effect, and this has altered the timing, nature, and size of the opportunity to transfer water allocations from the Goulburn to the Murray. This has meant that market participants have needed to adapt, with some (especially lower Murray irrigators) changing their water portfolio management strategies.

In January 2026, the River Murray in South Australia was declared by Minister Murray Watt as critically endangered, after recommendation from the Threatened Species Scientific Committee. This provided the ecosystems with some of the highest levels of legal protection under the Environment Protection and Biodiversity Conservation Act (EPBC Act). This has signalled heightened ecological risk at a whole-of-system level and can influence future land and water use decisions across the Basin (including in Victoria).

Such a designation typically increases regulatory scrutiny and strengthens the policy imperative to prioritise environmental outcomes, which may constrain approvals for new or intensive irrigation developments where additional water demand or altered flows could exacerbate downstream impacts. It may also accelerate consideration of further environmental water recovery, tighter

operational rules, or changes to water sharing arrangements, increasing planning and investment uncertainty for proponents of land use change.

1.4 What is horticultural headroom?

A key concept in this report is 'horticultural headroom'. This concept highlights the relative proportion of available water that could potentially be used by permanent horticulture plantings. As noted below, this does not mean that permanent horticulture plantings will actually receive the available water under any water availability scenario.

'Horticultural headroom' is the gap between the total volume of water available for consumptive use and the volume of water demanded (i.e., could be used) by permanent horticulture in a given year.

Presenting the volume of water available in terms of horticultural headroom offers insights into how market participants might behave when the relatively fixed demands of horticulture form a larger share of the available water. According to rational economic decision making, when water availability is limited and water allocation prices are high, more entitlement holders will choose to sell their water to higher-valued water users, where physical constraints and trade barriers permit. The key assumption in this analysis is that permanent horticulturalists generally place a higher value on water than those in broadacre cropping (e.g., cotton and rice) and dairy because watering is essential to maintain long-lived horticultural assets with inflexible water demands.

In practice, water market participants (e.g., dairy farmers or rice growers holding entitlements) may not opt to sell allocations for various reasons. The collective decisions of hundreds of individual water users, along with physical system or delivery constraints, influence how much water permanent horticulturalists can access at the required time and place.

Given the investment in long-lived permanent horticultural assets, prices are likely to rise notably when horticulturalists compete for limited water supplies. Consequently, there will be upward pressure on water allocation markets before the horticultural headroom is fully exhausted. The concept of horticultural headroom illustrates the risk of these market effects, enabling participants to plan and structure their decisions accordingly.

While analytically useful, the concept of horticultural headroom does not account for several key factors that will influence the timing and extent of market impacts, including:

- **Responses by permanent horticulturalists:**

  In response to high prices or water shortages, overall horticultural demand may decrease because of seasonal declines, permanent drying of planted areas across some or all orchards, earlier replanting or redevelopment plans, or if planned developments are postponed or abandoned. Notably, given the age profile of current almond plantings, there is potential for substantial replanting within the next decade.

- **Carryover of water from one year to the next:**

  Carryover plays a vital role in addressing water deficits, particularly at the start of each water year. Carryover was omitted from this analysis to emphasise the risk of water shortages during a period of multiple dry years, when carryover reserves are likely to be exhausted.

- **Trading of water allocations (inter-valley trade and across the Barmah Narrows in the Murray):**

  Intervalley trade and allocation trade across the Barmah Narrows in the Murray also play a crucial role in water availability dynamics across the southern connected system. The extent to which intervalley trade influences water availability in a given water year depends on complex market dynamics, so it was excluded from the quantitative analysis. This makes the analysis straightforward and directly comparable with the results in 2022 and 2025.

To calculate horticultural headroom, estimates of water availability and demand were generated using the calculations shown in **Box 1**.

Water usage estimates for 2024-25 serve as the starting point for the water demand analysis, as they are the most recent water year with complete data available across the three states. Therefore, they should not be treated as a definitive baseline. Instead, the 2024-25 demand estimate indicates the potential water demand under similar climatic and economic conditions. This method is useful for two reasons:

- First, it provides a detailed estimate of the age profile of current permanent horticulture plantings to forecast future water demand at full maturity.
- Secondly, it provides a simple method for annual monitoring of the balance between water availability and demand.

In the 2022 water availability and demand analysis, estimates of potential additional water demand from existing permanent plantings were expanded to include growth in citrus, grape, and almond crops, in line with the 2020 analysis. The most recent crop area data for the 2022 analysis show that between 2018 and 2021, almonds remained the dominant crop, with an expanding footprint, while areas planted to citrus and grape (dried, wine, and table) also increased. The 2025 SunRISE Mapping and Research report indicated that almonds and citrus continued to expand, whereas wine grape plantings declined. As a result, this analysis considers the change in crop size across almond, citrus, and grape (dried, wine, and table) plantings.

Box 1

Water demand estimates are calculated as follows:

• **2024-25 water demand estimate** = the sum of 2024-25 water usage estimated to be used by permanent horticulture, derived from data sources provided by the respective agencies.

• 2024-25 water demand estimate (full maturity) = **2024-25 water demand estimate** + estimate of water demand by existing permanent horticulture plantings at their full maturity.

• Projected water demand estimate (full maturity) = 2024-25 water demand estimate (full maturity) + estimate of water demand by known planned permanent horticulture plantings at their full maturity.

The 2024-25 water demand estimate (full maturity) and Projected water demand estimate (full maturity) assume no reductions in existing horticultural plantings.

Water availability estimates (or water available for irrigation) in five scenarios (wet, average, dry, moderately dry and extreme dry) are calculated as follows:

• **Total water available for irrigation** = ((Total entitlement on issue – environmental holdings – urban water holdings included in the consumptive pool) x (end of season allocations to each entitlement under each climatic scenario)) + allocations from non-irrigation entitlements made available to the market.

1.5 Data access and stakeholder engagement

Updated estimates of water availability and permanent horticultural demand were developed using the best available data from relevant government agencies and water authorities in Victoria, NSW, and SA. To promote transparency, Appendices B, C, and D detail all data sources and the data used to inform the analysis. To support a consolidated assessment of availability and demand in the connected Murray, a key part of this updated analysis involved targeted stakeholder engagement across the Basin states to identify the most reliable data sources and any improvements to data since the 2022 analysis. The cooperation and willingness of all stakeholders contacted to provide data were crucial to the success of the updated analysis presented in this report. Table 5 shows the stakeholder agencies from which data were obtained. Stakeholders reviewed a draft of the report and confirmed their support for the data included in the analysis. Reaching cross-jurisdictional agreement on availability and demand estimates is essential for effective discussions and decisions on water management at intergovernmental forums such as the Ministerial Council.

Table 5 Stakeholder agencies engaged to source the best available data

Jurisdiction	Stakeholders
Victoria	Department of Energy, Environment and Climate Action (Agriculture Victoria, Water and Catchments)
South Australia	Department for Environment and Water
New South Wales	Department of Climate Change, Energy, Environment and Water
Commonwealth	Department of Climate Change, Energy, Environment and Water (via the Commonwealth Environmental Water Holder)
Other agencies	Consultants, growers and industry experts with knowledge of agricultural development and water markets in the southern Murray-Darling Basin.

2. Results

The results are shown for two regions in the southern Murray-Darling Basin: the connected Murray and the lower Murray. Both regions exclude the Murrumbidgee and Lower Darling water systems.

All data, sources, and assumptions used in the analysis are included in the appendices. To help readers quickly interpret the results, a summary of the key data sources and assumptions is provided below.



Key analysis notes for the connected Murray and lower Murray

Water availability estimates

Water availability estimates are based on entitlement on issue data sourced from the respective state government departments. Consumptive water availability estimates are net of environmental water and urban water holdings, as well as conveyance licences in NSW (see Appendix B).

Water availability estimates take into account water allocations that are made available to the market from time to time (e.g., allocations to the Melbourne metropolitan water retailers' entitlements) (see Appendix B).

Water availability scenarios were derived from historical end-of-season allocations, as indicated throughout the analysis (see Appendix B).

Only within-region water availability is considered. WSP has assumed that no water allocations can be traded into the regions from connected zones, such as the Murrumbidgee for the connected Murray, and the Goulburn, the Murrumbidgee, and the upper Murray for the lower Murray. In relation to the Murrumbidgee, additional water could be traded into the lower Murray. However, WSP is aware of significant permanent horticultural developments in the Murrumbidgee that could reduce this source of water availability over time.

Water demand estimates

Water demand estimates were derived by:

Accessing 2024-25 actual water usage from the respective responsible entities. In some cases, assumptions were required to estimate usage by permanent horticulture. Data limitations prevented further disaggregation by crop type. In the NSW Sunraysia, demand from permanent horticulture could not be readily separated from demand by annual horticulture. All assumptions were derived in collaboration with the respective entities (see Appendix C).

Estimating the additional water demand from existing and projected permanent horticultural plantings based on the latest crop area data, as well as information from industry professionals (see Appendix D).

Estimates of the additional water demand from existing permanent plantings include recent almond, citrus, and grape (dried, wine, and table) plantings. Analysis of the best

available crop area data shows that between 2021 and 2024, almonds and citrus were the dominant crops with expanding footprints, while grapes decreased in area (SunRISE Mapping and Research 2025). This was mainly in the lower Murray, including the Edward-Wakool system and irrigated areas along the Murray River upstream to the Barmah Narrows. Future water usage was forecast based on water application rates (ML / ha) for:

- almond trees aged one to eight years provided by the Almond Board of Australia for the 2020 analysis.
- citrus trees aged one to seven years provided by industry professionals for the 2022 analysis.
- grape vines aged one to three years provided by industry professionals for the 2022 analysis (see Appendix D).

Any reductions in planting area were assessed under the assumption that any trees removed were already at full maturity. Accordingly, the associated calculations treated the resulting reduction in water demand as equivalent to the demand at full maturity of those trees.

Estimates for the maximum water usage of projected permanent plantings were derived from information (size of developments and water application rates) provided by industry professionals in each of the three states. Based on known developments for which industry professionals had a reasonable level of confidence that would occur in the next three to five years, an estimate of the maximum water usage was derived. Estimates for projected demand include planned developments for a wide range of permanent horticultural crops, including almonds, citrus, wine grapes, and table grapes.

The horticultural headroom estimates do not take into account:

- Any potential adjustments to other crops that may occur due to water availability and commodity markets.
- Any potential adjustment strategies that reduce irrigation application rates or dry-off old orchards and underperforming blocks during times of severe drought or low water availability.
- The likelihood is that not all known permanent horticultural developments at various stages of development will go ahead.
- Land use changes from crops with lower water use to crops with higher water use. For example, conversions from wine and dried grapes to table grapes and citrus. Such conversions could already be underway and could be considered in future updates of the analysis.

Key analysis notes for the lower Murray only

WSP has assumed that no groundwater is available in the Lower Murray region from the Lower Murray Groundwater Aquifer. This approach is consistent with the 2022 analysis. This was confirmed in Victoria. Similarly, for NSW, a map of extraction sites was accessed for the 2020 analysis. For practical purposes, almost all extraction sites

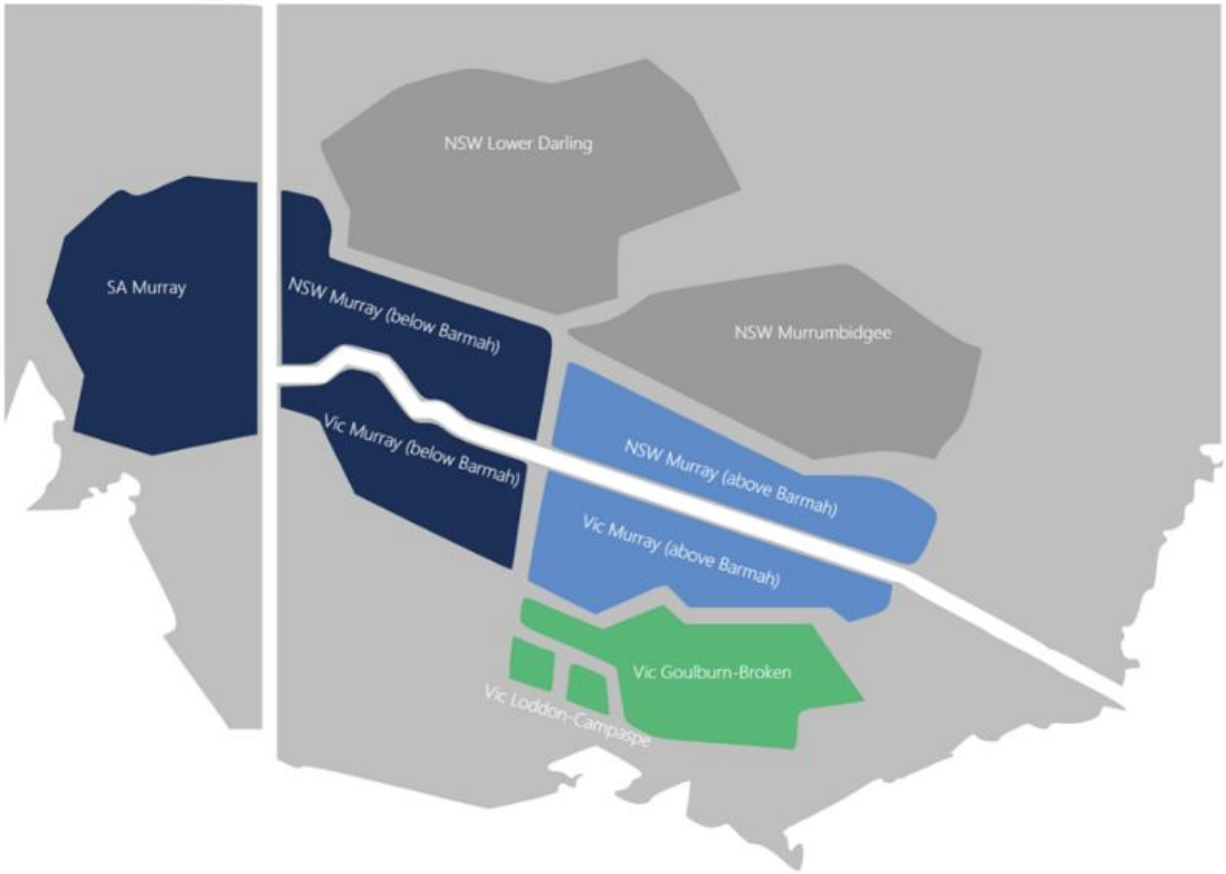
were within the Murray Irrigation Limited district. This suggests that most groundwater users who also have access to surface water would access water from above the Barmah Narrows. It also suggests that groundwater users in the NSW Murray are unlikely to be permanent horticulturalists. In addition, there are known groundwater quality issues in the Lower Murray Groundwater Aquifer that are actively managed, including by salt interception schemes to lower the water table and prevent it from connecting to the river. Therefore, for the lower Murray region, water availability is limited to surface water, as we assume no groundwater is available.

2.1 Horticultural headroom in the connected Murray

The **connected Murray** region is defined as all major water systems within the southern Murray-Darling Basin with connectivity to the Murray River, **excluding the NSW Murrumbidgee and the Lower Darling** (Figure 9).

The specific water zones captured within the connected Murray region are: Vic 6 Murray, Vic 7 Murray, Vic 1A Greater Goulburn, Vic 2 Broken, Vic 3 Lower Goulburn, Vic Campaspe, Vic Loddon, Vic Lower Broken Creek, northern Victorian groundwater (Katunga, Shepparton and Lower Campaspe Valley), NSW 10 Murray, NSW 11 Murray, NSW lower Murray groundwater, and South Australian Murray.

Figure 9 Water systems in the connected Murray



2.1.1 Water availability and demand in the connected Murray; Existing permanent horticulture (full maturity)

Based on actual water usage data for 2024-25, the estimate of current water use by existing permanent horticulture in the connected Murray is 1,239 GL.[9] This reflects a 111 GL increase from the estimate provided in the 2022 analysis and is driven by a net increase in plantings since the previous version's analysis (see details in Section 3).

Across the connected Murray, there are significant areas of permanent horticultural plantings that have yet to reach maximum water use. These plantings, established between 2021 and 2025, cover an estimated 21,000 hectares of permanent horticultural areas across all states. However, this figure is offset by 4,000 hectares of grape plantings that have been removed, which may take time to be replaced with alternative permanent crops, as many winegrape growers lack the necessary funds to transition quickly. (Appendix D - Table 54). The estimated age profile of the immature permanent horticulture crops and their associated water application rates by year include:

- Almonds, with a water profile provided by the Almond Board of Australia during the 2020 analysis (up to 14 ML per hectare at maturity).
- Citrus, with a water use profile provided by industry professionals during the 2022 analysis (up to irrigation of 12 ML per hectare at maturity).
- Grapes (table, wine, and dried), with water use profiles provided by industry professionals during the 2022 analysis (up to irrigation 10 ML per hectare at maturity) (see Appendix D).[10]

Based on estimates of the age profiles of permanent plantings in the ground, our estimate of the additional water demand of these plantings at maximum water use is 67 GL. This suggests that, in the future, the total water demand for existing permanent horticulture in the connected Murray will be 1,306 GL per year, assuming all other factors remain constant (i.e., no demand response in the permanent horticultural sector). This represents an increase of 32 GL compared with the estimate in the 2022 analysis.

Figure 10 plots several possible future water availability scenarios for the connected Murray Region against the best estimate of future water demand for all existing permanent horticulture in the region. Water availability estimates are shown for:

- surface water and groundwater
- surface water without groundwater

Generally, groundwater sources in the connected Murray region either do not overlap geographically with existing permanent horticulture demand (in the lower Murray) or, if they do, are not of sufficient quality to be used directly for irrigated agriculture. Therefore, to present a comparative, worst-case water availability scenario for the connected Murray, we have included a water availability estimate that excludes groundwater.

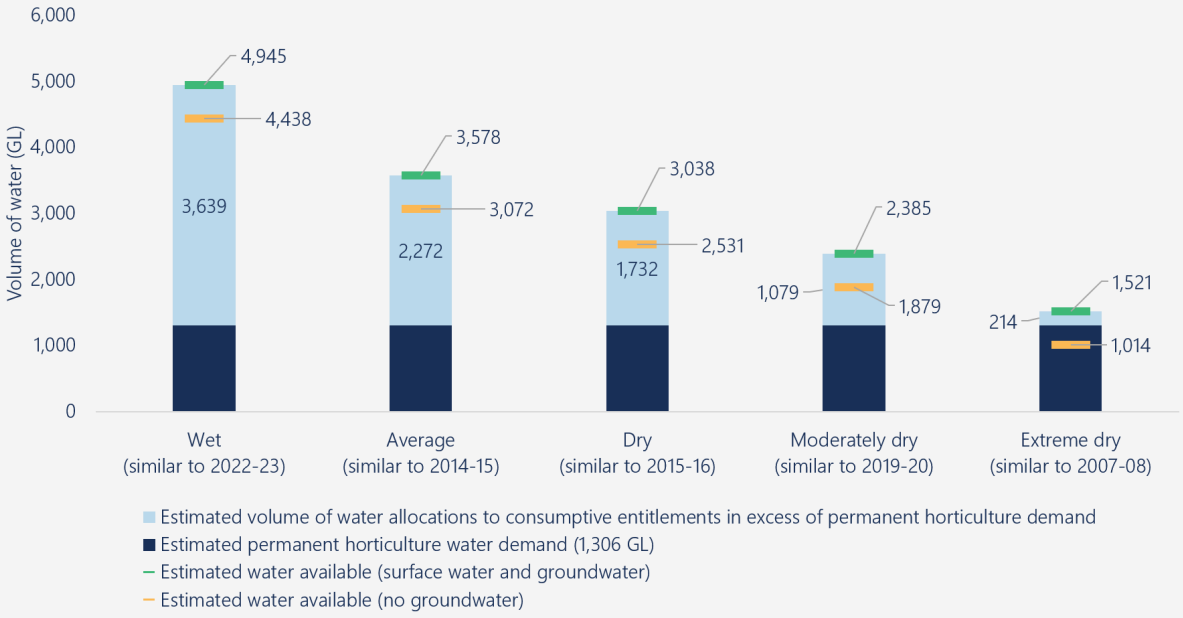
Key messages

[9] As noted above, disaggregating water usage data by crop type was challenging. The 2024-25 estimate for water by permanent horticulture likely includes some annual horticulture demand in NSW

[10] Water application rates vary across dried, wine, and table grapes. However, due to data limitations an average water application rate was applied to all existing grape crops and potential grape developments.

- The 2025 analysis confirms previous findings that the current fixed water demands of permanent horticulture industries in the connected Murray are increasing. However, it also shows that the rate of new plantings is lower than in the last report, leading to slower forecasted growth in water demand by permanent horticulture.
- During future periods of extreme dry water availability, when there's no carryover from the previous year or an allocation trade from the Murrumbidgee, water demand from existing permanent horticulture plantings (at full maturity) in the connected Murray may be approximately 214 GL lower than the available water (estimated demand of 1,306 GL compared with estimated surface and groundwater availability of 1,521 GL).
- If groundwater (approximately 500 GL of entitlement on issue across the connected Murray) is removed from the extreme dry scenario, the estimated water needs of existing permanent horticulture (at full maturity) amount to 129 per cent of the total consumptive water available in the connected Murray region. In other words, under this scenario, demand from existing permanent horticulture may exceed the total volume of water available for consumptive use by around 292 GL (estimated demand of 1,306 GL compared to estimated surface water availability of 1,014 GL).
- Under moderately dry conditions, similar to the 2019-20 drought, it is estimated that demand from permanent horticultural plantings at full maturity will be 55 per cent of estimated water availability (1,306 GL compared with 2,385 GL of surface and groundwater availability). This leaves a horticultural headroom of 1,079 GL. Removing the approximately 500 GL of groundwater entitlement on issue reduces this horticultural headroom to 573 GL.

Figure 10 Water availability scenarios and existing permanent horticulture water demand (permanent plantings at full maturity), connected Murray region



2.1.2 Water availability and demand in the connected Murray: Projected permanent horticulture at full maturity

For this analysis, data on potential horticultural developments at various planning stages were gathered from government sources and industry professionals with local knowledge of agricultural developments. The data included both potential increases and decreases in horticultural plantings.

It must be noted that there is significant uncertainty about future permanent horticultural developments. This is because various climatic and economic factors will influence enterprises' decisions on whether to proceed with planned projects. These factors include seasonal conditions, such as consecutive dry years, as well as fluctuating commodity prices, labour shortages, and the availability and cost of capital. In summary, not all projects identified during data collection will necessarily proceed. Additionally, it remains possible that not all plantings have been documented by the sources.

It is estimated that an additional 87 GL could be required for permanent horticulture plantings (at full maturity) that are not yet in the ground. This comprises:

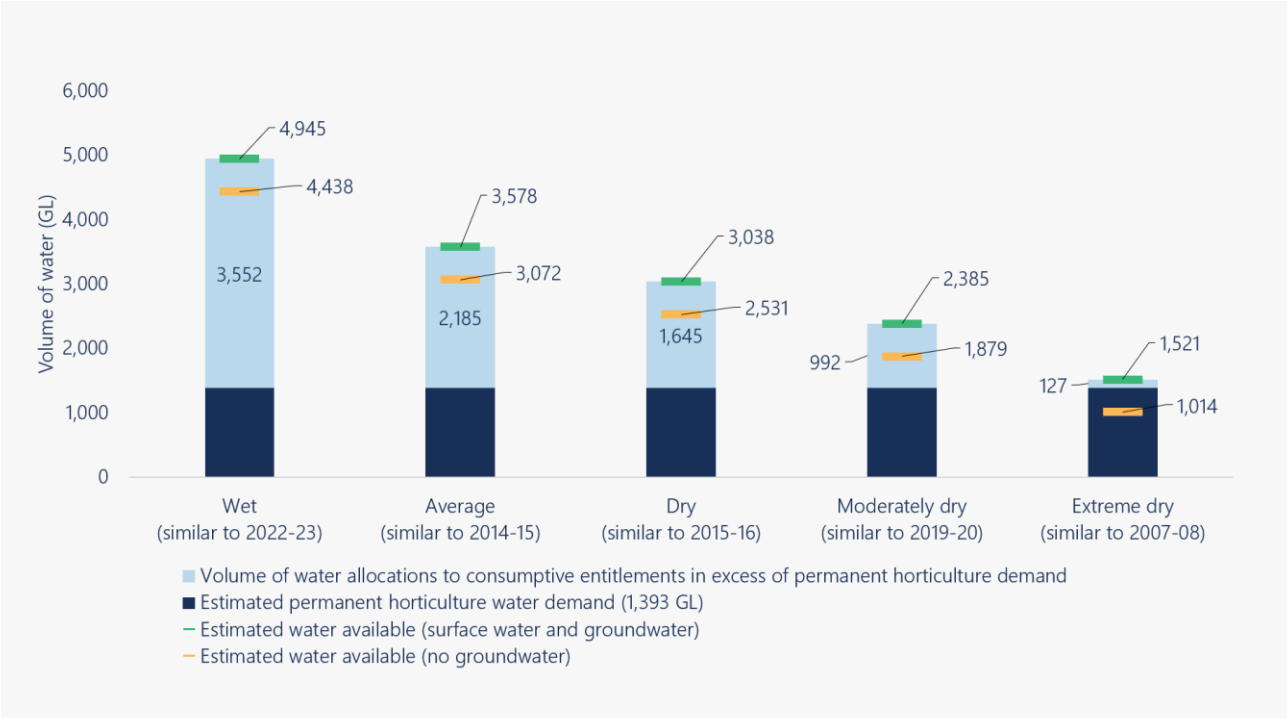
- Potential almond developments = 90 GL
- Potential citrus developments = 16 GL
- Potential grape developments (reductions) = -19 GL

Figure 11 plots several possible future water availability scenarios in the connected Murray region against the best estimate of the future water demands of all existing projected permanent horticulture in the region (at full maturity).

Key messages

- The findings of the 2025 analysis indicate that under an extreme dry scenario with groundwater excluded, water demand from existing and projected permanent horticulture plantings will exceed the available water. This changes when groundwater is included, with just about enough water available.
- During periods of future extreme dry water availability (with no carryover from the previous year, and no allocation trade in from the Murrumbidgee), water demand will be 127 GL lower than water availability (estimated demand of 1,393 GL compared to estimated availability of 1,521 GL) (Figure 11).
- If groundwater (approximately 500 GL of entitlement on issue across the connected Murray) is removed from the extreme dry scenario, the estimated water requirements of existing and projected permanent horticulture (at full maturity) are approximately 137 per cent of the estimated total consumptive water availability in the region, representing a water availability deficit of approximately 379 GL (Figure 11).

Figure 11 Water availability scenarios and projected permanent horticulture water demand, connected Murray region

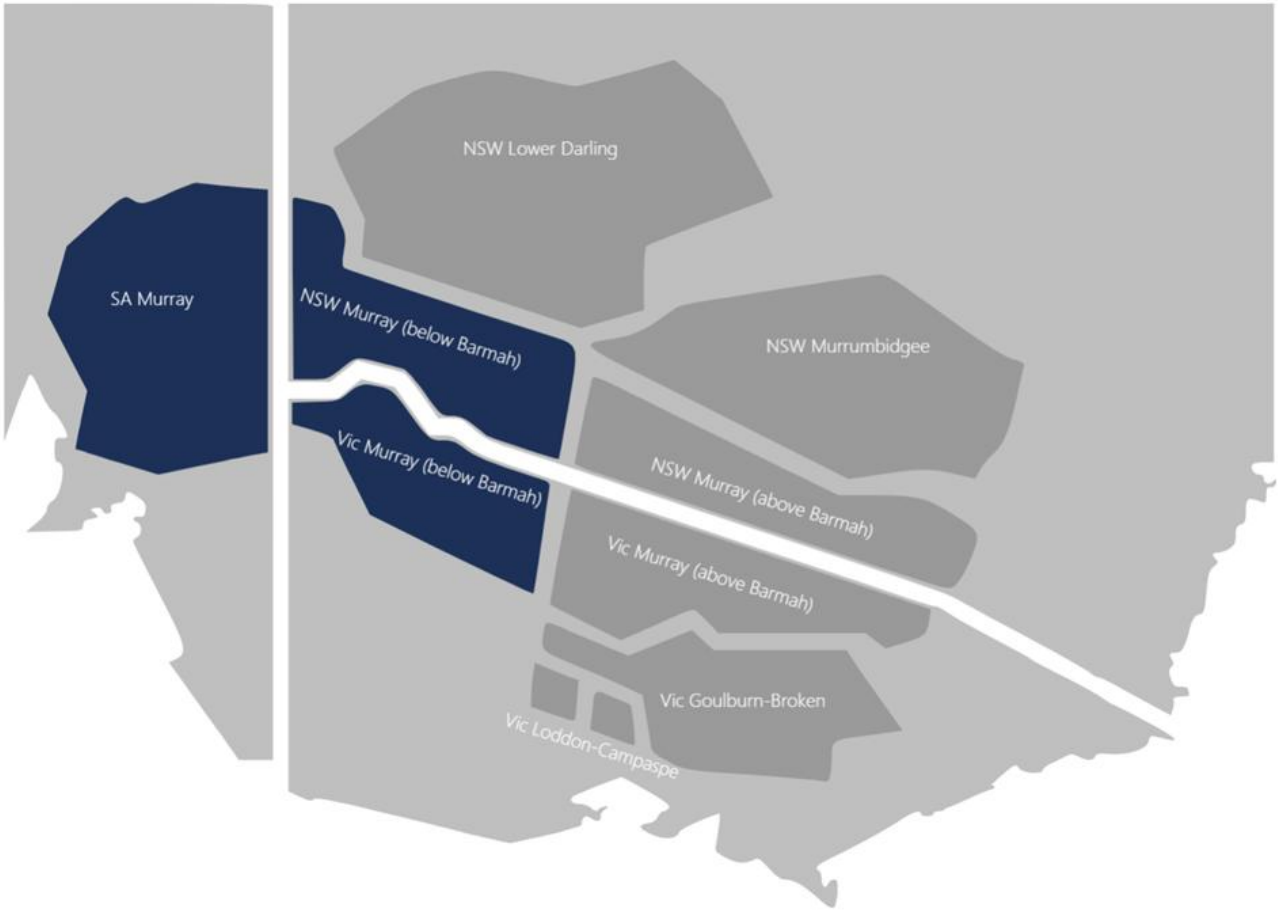


2.2 Horticultural headroom in the lower Murray

The lower Murray region includes all water systems within the connected Murray downstream of the Barmah Narrows, excluding the Lower Darling and the Murrumbidgee (Figure 12). The specific water systems captured within the lower Murray region are: Vic 7 Murray, NSW 11 Murray, and South Australian Murray.

Similar to the 2022 analysis, results for this subregion are presented separately because the lower Murray includes the Victorian Sunraysia, New South Wales Sunraysia, and South Australian Riverland, areas with a high concentration of permanent horticulture. There are also growing concerns about pressure on water resources in the lower Murray compared with the connected Murray. These concerns are mainly driven by trade restrictions that limit the ability to trade water into the lower Murray from above the Barmah Narrows and out of the Goulburn and Murrumbidgee valleys. Notably, the timing, nature, and extent of opportunities to transfer water allocations into the lower Murray have evolved over time. The most significant development since the previous report is that more than three years have passed since the new Goulburn to Murray Trade Rule was implemented, providing market participants sufficient time to adapt their water management strategies.

Figure 12 Water systems in the lower Murray



2.2.1 Water availability and demand in the lower Murray: Existing permanent horticulture at full maturity

Based on actual water-use data for 2024-25, our estimate of current water demand from existing permanent horticulture in the lower Murray is 1,132 GL per annum. This is an increase of 83 GL compared with the estimate provided in the 2022 analysis, explained by a net increase in plantings since the previous version's analysis (see details in Section 3).

When accounting for the maturity profile of recent permanent horticulture plantings, water demand from existing permanent horticulture at full maturity in the lower Murray region could be 1,199 GL. This is an increase of only 3 GL compared with the estimate provided in the 2022 analysis, explained by the large net reduction in new plantings since the 2022 analysis. (see details in Section 3).

This is based on the estimated age profile of immature permanent horticulture crops and their associated water application rates by year, including:

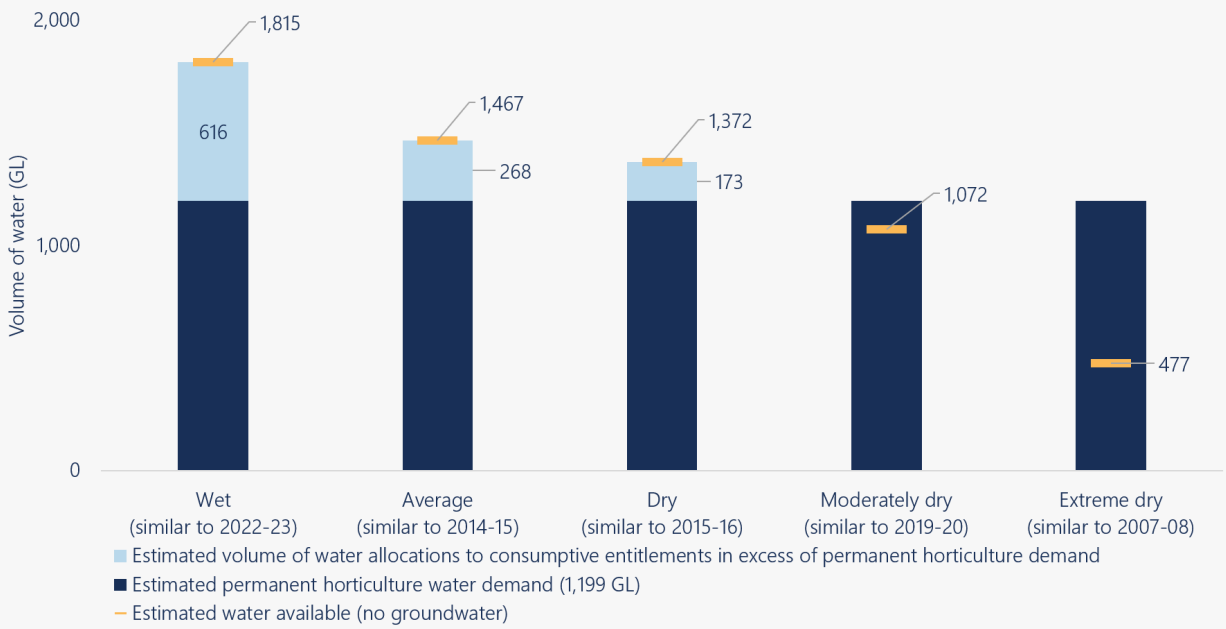
- Almonds, with a water use profile provided by the Almond Board of Australia during the 2020 analysis (up to 14 ML per hectare at maturity).
- Citrus, with a water use profile provided by industry professionals during the 2022 analysis (up to 12 ML per hectare at maturity).
- Grapes (table, wine, and dried), with water use profiles provided by industry professionals during the 2022 analysis (up to 10 ML per hectare at maturity) (see Appendix D).[10]

Lower Murray permanent horticultural demand (at full maturity) is approximately 92 per cent of the estimated permanent horticultural demand across the connected Murray region. Consistent with the analysis presented for the connected Murray above, Figure 13 plots several possible future water availability scenarios in the lower Murray region against the estimate of future water demand for all existing permanent horticulture in the region.

Key messages

- The 2025 analysis reinforces the previous findings that in moderately dry and extreme dry years, there will likely be no horticultural headroom available in the lower Murray.
- Whilst the analysis finds that the current fixed water demands of permanent horticulture industries in the lower Murray are increasing, the rate of increase has slowed compared to previous versions.
- In an extreme dry year, water demand from existing permanent horticulture in the lower Murray at full maturity may exceed availability by approximately two and a half times, consistent with the 2022 analysis. In the 2025 analysis, estimated demand was 1,199 GL, while estimated availability was 477 GL. This means there will likely be significantly increased competition for water, putting upward pressure on prices.
- In a moderately dry year – similar to 2019-20 – within-region availability may be sufficient to meet only 89 per cent of water demand from existing permanent horticulture in the lower Murray at full maturity (estimated demand of 1,199 GL vs estimated availability of 1,072 GL).
- As highlighted in the 2022 report, in the absence of a demand-side response, this deficit in within-region available water will need to be met through carryover and trade of water into the lower Murray. While this is possible, it is not guaranteed. There will likely be increased competition among water users to secure the volumes they demand, either by purchasing entitlements or bidding for allocations on the spot market. In turn, this may mean that water users who do not already own entitlements may not be able to secure the water they demand on the allocation market or may pay higher prices to access allocation water. This is likely to have similar implications for water market participants in connected trading zones.
- Even under an average climatic scenario, the volume of water available for consumptive use in the lower Murray may be only 22 per cent higher than permanent horticulture demand. However, under a wet scenario, there is sufficient water available (51% more than demand) – noting that this is driven by how the wet scenario has been modelled in this report (see details in Section 3).

Figure 13 Water availability scenarios and existing permanent horticulture water demand (permanent plantings at full maturity), lower Murray region



Note: For the lower Murray, 'estimated water available (surface water and groundwater)' is not shown because there is no groundwater available in the lower Murray region from the Lower Murray Groundwater Aquifer. See pp. 14-15 for further details.

2.2.2 Water availability and demand in the lower Murray: Projected permanent horticulture at full maturity

For this analysis, data on potential horticultural developments at various stages of planning were gathered from a range of government sources and industry professionals with on-the-ground understanding of agricultural developments. The data included both potential increases and decreases in horticultural plantings.

However, there is considerable uncertainty about future permanent horticultural developments. This is because a range of climatic and economic factors will influence an enterprise's decision to proceed with a planned development. These drivers include seasonal conditions, such as successive dry years, as well as changing commodity prices, labour shortages, and the availability and cost of capital. In summary, it is highly likely that not all the developments identified during the data gathering will proceed. Additionally, it is possible that not all plantings have been captured by the sources.

In the 2022 analysis, all planned development data focused on the lower Murray, and as such, the analysis for the connected Murray and lower Murray was the same. In the 2025 analysis, some planned development data included developments in the upper Victorian Murray, meaning that in this report, planned developments are no longer identical between the connected Murray and the lower Murray.

It is estimated that an additional 45 GL could be required by permanent horticultural plantings (at full maturity) not yet in the ground. This comprises:

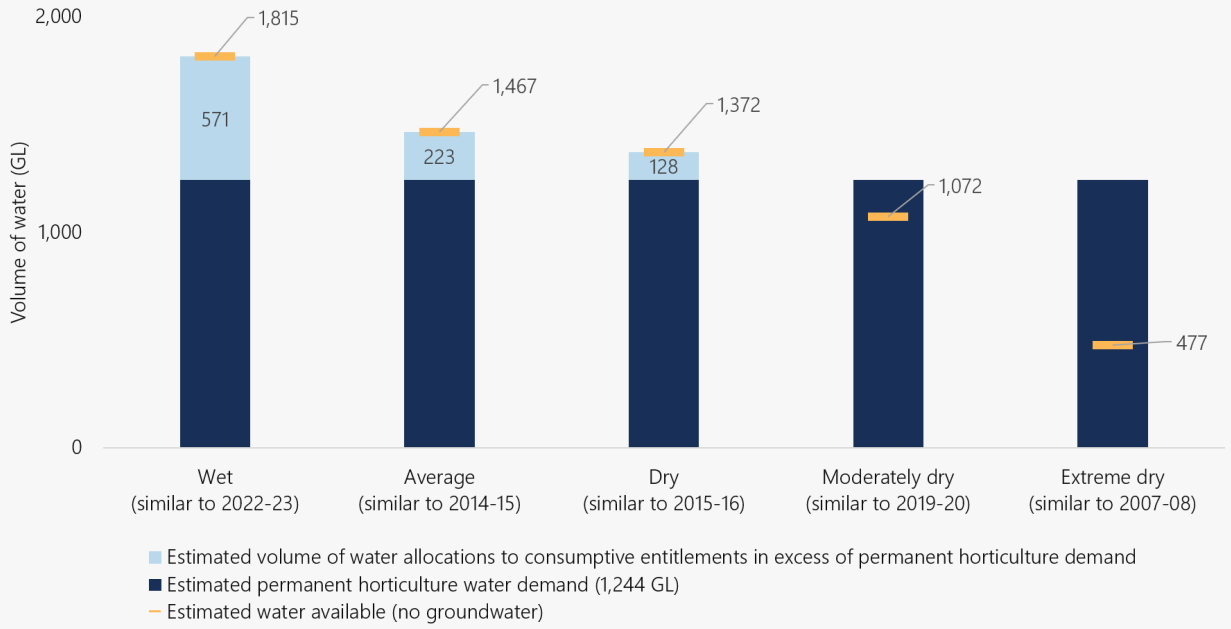
- Potential almond developments = 48 GL
- Potential citrus developments = 16 GL
- Potential grape developments = -19 GL

Figure 14 plots several possible future water availability scenarios in the lower Murray region against the best estimate of the future water demands for all existing and projected permanent horticulture in the region (at full maturity).

Key messages

- Consistent with the previous report, applying projected permanent horticultural demand to the water availability scenarios continues to indicate a concerning outlook for the lower Murray relative to the connected Murray. This is driven by the combination of concentrated permanent demand in the lower Murray region and a lower estimate of availability in this region (i.e., total water availability in the connected Murray will not be available to permanent horticulture in the lower Murray).
- In extreme and moderately dry years without carryover or intervalley trade into the lower Murray, there is likely to be no horticultural headroom available (Figure 14).
- In a moderately dry year, demand from permanent horticulture may outstrip availability by approximately 16 per cent: estimated demand of 1,244 GL and estimated surface water availability of 1,072 GL (Figure 14)
- Such findings continue to highlight to market participants the critical importance of access to the broader connected southern Murray-Darling Basin water market (through allocation trade from the Goulburn, Murrumbidgee and upper Murray) and carryover.

Figure 14 Water availability scenarios and projected permanent horticulture water demand including all developments, lower Murray region



3. Comparison of 2022 and 2025 results

One of the benefits of using a consistent method in analysing water availability and demand is that the results can be compared over time. Comparable data and analyses are important from two perspectives.

First, they support decision-making by water market participants and stakeholders, including:

- Entitlement holders and water users across all irrigated agricultural industries in the southern Murray-Darling Basin are making decisions about water availability and pricing expectations.
- New water market entrants and existing horticulturalists who may be looking to expand their enterprises within the southern Murray-Darling Basin, and who need to estimate water costs and availability, as well as the enterprise risks associated with water access.
- Government decision-makers, committees, and policy makers responsible for further water market reforms.

Second, there is no centralised source of water availability and demand data. As set out in the appendices, generating estimates of water availability and demand requires multiple data sources, and assumptions are often needed to estimate water use by permanent horticulture. Thus, it is important to understand the extent to which improved data sources may be influencing the results, rather than any material changes in water availability and demand.

WSP compared the results of this analysis with those from 2022 to demonstrate the order of magnitude of any differences and their drivers. The differences between the 2022 and 2025 results are set out as follows:

- Comparison of modelling approaches in estimating water availability and demand
- Comparison of estimates of water available for irrigation
- Comparison of estimates of baseline water use by permanent horticulture
- Comparison of the estimates of water demand by existing permanent horticulture plantings at full maturity
- Comparison of estimates of projected water demand from planned horticulture plantings

3.1 Comparison of estimates of water available for irrigation

In 2025, the water availability estimates for the connected and lower Murray are higher than those from the 2022 analysis (Table 6). In 2025, the water availability estimates for the connected and lower Murray are higher than those from the 2022 analysis (Table 6). The key drivers of these differences are:

Changes to entitlement on issue available for consumptive use in 2025.

- Between 2022 and 2025, the entitlement on issue available for consumptive use in Victoria increased by approximately 71 GL, predominantly due to the allocation of shares issued under the Irrigators' Share Distribution as analysed in the 2022 report (Victorian Goulburn high and low reliability). However, in NSW, EOI for consumptive use decreased by approximately 40 GL, predominantly due to government purchases of environmental water – especially in the NSW Murray and Murrumbidgee.
- The net increase resulting from the above means the total pool of consumptive entitlements to which allocations can be made in the connected Murray and lower Murray was higher in the 2025 analysis.

Change in reference year of the 'wet' scenario from 2011-12 (2022 report) to 2022-23 (2025 report).

- Since the 2022 report was released in 2023, the southern Murray-Darling Basin has experienced an extended three-year La Niña period, driving record rainfall in the region. Allocations to entitlements were higher than in 2011-12, mainly driven by allocations to low reliability entitlements, which are rarely allocated water. As such, the reference 'wet' year was changed to 2022-23, when rainfall was at its highest. This reflects a more representative wet scenario for water market participants by drawing on the latest information available.
- None of the other climate scenarios were changed, nor were they affected by this modification.
- Total water allocated to entitlements for consumptive use (GL) was higher in 2022-23 than in 2011-12 by 883 GL in the connected Murray and 229 GL in the lower Murray. These differences account for a considerable proportion of the increase in the total estimate of water available to irrigators compared with 2022. This was driven by 100% allocations to low-reliability entitlements in the Goulburn and Murray, compared with 0% in 2011-12 (providing an extra 723 GL).

Potential water allocation made available to the market.

- As part of the previous change in reference year, WSP also examined the assumptions used to determine water available from water corporations and the environment. This was previously done by examining data from 2013-14, 2016-17 and 2021-22. It has now been changed to refer to data from 2021-22, 2022-23 and 2023-24, which are more recent and representative wet years. More information on this is in Appendix B.
- Environmental water holder and water corporation trading activities likely depend on many factors, not just the year's climate. For example, the climate of previous years can influence the volume of water available for trading. The health of various water sites can also influence the environmental water holder's priorities and their decision to trade (or not to). Broader business strategies and approaches can also change over time, influencing trading decisions.
- During the 2022 analysis, the years (under a wet scenario) used to calculate an average water volume yielded 46 GL in the upper Murray / Goulburn and 114 GL in the lower Murray. Using updated 2025 data, the average was 48 GL in the upper Murray / Goulburn and 135 GL in the lower Murray, increasing overall water availability in a wet year.

Table 6 Comparison of water availability estimates generated in 2022 and 2025

	Connected Murray			Lower Murray		
	2020-21	2024-25	Difference	2020-21	2024-25	Difference
Wet (similar to 2022-23)	4,070 GL	4,945 GL	+ 875 GL	1,594 GL	1,815 GL	+ 221 GL
Average (similar to 2014-15)	3,552 GL	3,578 GL	+ 26 GL	1,460 GL	1,467 GL	+ 7 GL
Dry (similar to 2015-16)	3,007 GL	3,038 GL	+ 31 GL	1,364 GL	1,372 GL	+ 8 GL
Moderately Dry (similar to 2019-20)	2,359 GL	2,385 GL	+ 26 GL	1,067 GL	1,072 GL	+ 5 GL
Extreme dry (similar to 2007-08)	1,502 GL	1,521 GL	+ 19 GL	473 GL	477 GL	+ 4 GL

3.2 Comparison of estimates of baseline water use by permanent horticulture

The baseline water demand estimates for permanent horticulture are slightly higher in 2025 than in 2022. These differences are shown in Table 7, and the key drivers are explained below.

Table 7 Comparison of baseline permanent horticulture demand estimates generated in 2022 and 2025

	Connected Murray			Lower Murray		
	2020-21	**2024-25**	**Difference**	**2020-21**	**2024-25**	**Difference**
Water demand estimate	1,128 GL	1,239 GL	+ 111 GL	1,049 GL	1,132 GL	+ 83 GL

Baseline years were updated to use the most recent water year for which data are available across the states.

- The 2022 analysis used 2020-21 as the baseline year, a water year characterised by relatively cool conditions, particularly during the summer peak irrigation period. This means permanent horticulturalists were less reliant on irrigation water and likely had low water application rates.

- The 2025 analysis uses 2024-25 as the baseline year, which was a water year characterised by relatively dry conditions and low rainfall. This means that, compared with 2020-21, water application rates by permanent horticulturalists were likely higher in 2024-25.

Increases in the commodity prices, particularly for almonds

- The price per kilogram of almonds has steadily increased from approximately $7/kg in December 2022 to just under $12/kg in June 2025, its highest level since 2015-16.[11,12]
- Additionally, global demand for almonds has grown steadily over the past 15 years, rising from approximately 0.8 million tonnes in 2010 to approximately 1.6 million tonnes in 2025. This price trend and longer-term demand growth have encouraged growers to plant more almonds, resulting in an additional 3,600 ha of almond trees planted since 2021
- The additional estimated water use for almonds outweighs any reduction in use from lower wine grape plantings, thereby increasing water demand across the Basin.

3.3 Comparison of the estimates of water demand by existing permanent horticulture plantings at full maturity

The water demand estimates for existing permanent horticulture plantings at full maturity are also slightly lower in 2025 than in 2022. These differences are shown in Table 8, and the key drivers are explained below

Table 8 Comparison of existing permanent horticulture demand estimates (at full maturity) with the 2022 analysis

	Connected Murray			Lower Murray		
	2020-21	2024-25	Difference	2020-21	2024-25	Difference
Water demand estimate (full maturity)	1,275 GL	1,306 GL	+ 31 GL	1,196 GL	1,199 GL	+ 3 GL
Estimate of additional water use by existing plantings at full maturity	147 GL	67 GL	- 80 GL	147 GL	67 GL	- 80 GL

[11] Almond Board of Australia, 2024, Almond Insights 2024–25, https://viewer.joomag.com/2024-25-aba-almond-insights/0387963001763097522/p20?short=
[12] AU$/kg of Nonpareil Extra No. 1 grade almonds.

3.3.1 Key improvements in estimating water demand

The estimates of existing permanent horticulture at full maturity in the connected Murray and lower Murray rely on combining a baseline demand estimate with an estimate of the additional water demanded by existing permanent plantings at full maturity. This second component of the demand calculation is key, as permanent horticulture plantings typically require higher water application rates as they age, peaking at full maturity.

Figure 15 compares the additional water demand from existing permanent plantings included in the 2022 and 2025 analyses. The drivers of the differences are as follows:

Additional water use by existing almond plantings

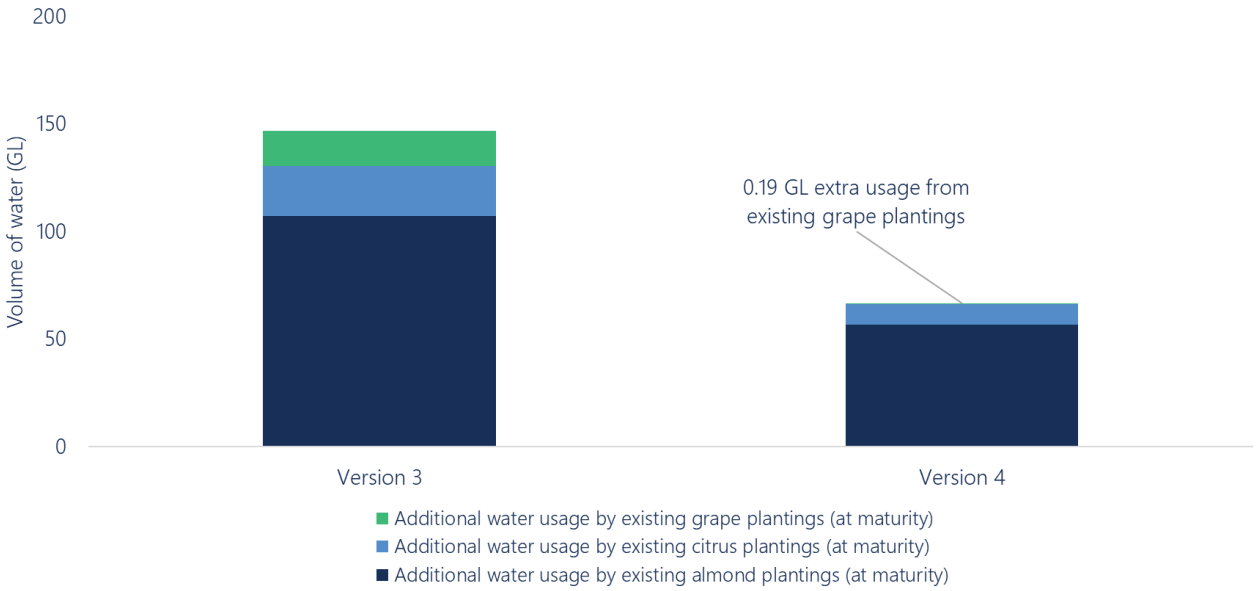
- The 2022 analysis estimated that an additional 107 GL of water would be demanded by existing almond plantings at full maturity. In contrast, the 2025 analysis estimated that only 57 GL of additional water would be demanded by existing almond plantings at full maturity.
- A key driver of this slowdown in water use from existing plantings is likely the continued maturation of many plantings initially established between 2017 and 2020. Water use from such plantings tends to increase rapidly in the first three years, then plateaus and remains relatively steady for the remaining life of the trees.
- Several orchards planted between 2007 and 2010[13] are approaching the end of their productive life and may be replanted in the coming years, according to industry sources. This could lead to short-term reductions in future water use from such plantings, followed by a return to their original use levels at full maturity.

Additional water use from other permanent crops

- The 2022 analysis estimated 39 GL of additional water demand from existing citrus and grape plantings (table, wine and dried). In 2025, existing citrus plantings are projected to use an additional 10 GL at maturity. While there was a net increase of 0.19 GL in additional water use by existing grape plantings at maturity, this was influenced by a mix of drivers. Declines were seen in table and wine grape plantings across all three states between 2021 and 2024 (3,285 ha). However, increases were also seen in table grapes (900 ha), which offset some of the above decline.
- The reduced water use from the removal of table and wine grape plantings between 2021 and 2024 was factored into 2024-25 and future water use. However, the increase in table grape plantings and their ongoing maturation helped drive the 0.19 GL increase above.
- The additional water demand for existing plantings (almond, citrus, and grapes) at full maturity was estimated to be 67 GL in 2025, lower than the 147 GL estimated in 2022.

[13] Australian Almonds, 2025. Almond Insights 2024/2025. Accessible via: https://viewer.joomag.com/2024-25-aba-almond-insights/0387963001763097522/p1?short=

Figure 15 Comparison of additional water usage of existing permanent horticultural crops at maturity included in the 2022 (baseline 2020-21) and 2025 (baseline 2024-25) analyses



3.4 Comparison of estimates of projected water demand from planned permanent horticulture plantings

3.4.1 Summary

The 2022 analysis of planned plantings showed a large increase in planned planting area across all three states for all three crop types, with a total estimated increase in water demand (at full maturity) of 175–409 GL. The 2025 analysis shows a reduction in this growth, with a combined increase in water demand for all three crop types across all three states (at full maturity) of 87 GL (noting that grapes are expected to decline by 19 GL – a net 1,899-hectare decline). Almond and citrus plantings continue to grow but at a reduced rate from the 2022 analysis, while wine grapes are expected to decrease in planting area, a reversal from the expected growth shown in the 2022 analysis.

3.4.2 2025 projected water demand from planned permanent horticulture plantings

Estimates of projected water demand from permanent horticultural plantings at various stages of planning (at full maturity) in the lower Murray and the connected Murray rely on data from industry professionals with on-the-ground knowledge in each of the three states. Based on known developments or reductions in plantings with a high likelihood of occurring over the next three to five years, future water demand estimates for planned plantings were developed.

The estimates show a 6,452-hectare increase in almond planting area, with 3,000 hectares in the upper Victorian Murray, 452 hectares in the lower Victorian Murray, and 3,000 hectares in the lower NSW Murray. It also shows a 500-hectare increase in citrus planting area in the lower NSW Murray and a 1,361-hectare increase in table grape planting area in the lower Victorian Murray. However,

the total increase in horticultural planting area is significantly reduced by large reductions in wine grapes, with an estimated 4,000 ha that could be removed across the lower Murray.

Reductions in wine grape plantings stem from a sharp fall in prices in recent years due to systemic global oversupply (down to $150-160/ton from historical $400-$480/ton for red wine varieties)[14], coupled with strong returns from alternative crops. As a result, 72% of wine grape operations reported losses in the 2023-24 season.[14] This downturn in wine grape production has resulted in an estimated ~1,900 hectare decrease in planting area across all three states, with South Australia modelled to drive the largest decline at an estimated 2,561 hectares.

Discussions with industry experts indicate a continued negative outlook for wine grape growers, with an estimated 19 GL reduction in water use in the lower Murray. Following years of economic and trade headwinds, more plantings are expected to be removed. Many businesses are focused on sustaining their current operations and financial performance, in some cases effectively on 'life support'. Many are also looking to convert their land to residential or lifestyle uses. Some growers are also looking to remove their grape plantings while investigating opportunities to convert to other crops. However, many of these properties will need additional infrastructure installed to support new crop types.

In January 2026, the River Murray in South Australia was declared critically endangered by Minister Watt, following a recommendation from the Threatened Species Scientific Committee and providing the ecosystems with some of the highest levels of legal protection under the EPBC Act. This has signalled heightened ecological risk at a whole-of-system level and is likely to influence future land and water use decisions across the Basin, including in Victoria.

Such a designation typically increases regulatory scrutiny and strengthens the policy imperative to prioritise environmental outcomes, which may constrain approvals for new or intensive irrigation developments where additional water demand or altered flows could exacerbate downstream impacts. It may also accelerate consideration of further environmental water recovery, tighter operational rules, or changes to water sharing arrangements, increasing planning and investment uncertainty for proponents of land use change.

3.4.3 Comparison between 2022 and 2025

Unlike the 2025 analysis, the 2022 analysis used two scenarios: Scenario A included all planned plantings with a high likelihood of proceeding under 1,000 hectares, and Scenario B included all planned plantings with a high likelihood of proceeding irrespective of size. At the time in 2022, many planned plantings in the region were being considered of varying sizes – leading to developing the two scenarios above. However, in 2025, general market sentiment drove a reduction in planned plantings, with particular focus on reductions in wine grape plantings. As a result, WSP decided to change its 2025 approach to projecting future new plantings/removals.

In this report, WSP compares Scenario B from the 2022 analysis with the 2025 analysis.

While Scenario B (in 2022) includes an additional 8 potential developments of 1,000 ha or larger, these developments are very large and inflate the estimate of projected permanent horticulture demand. Developments of this size already exist in the lower Murray (as noted in the 2022 report).

[14] CCW Cooperative, 2025, Riverland Transformation Pathways; a Comprehensive Industry Assessment and Strategic Framework.

Scenario B for 2022 showed large, planned increases in all three crop types across all three states: almonds by 16,039 hectares[15], citrus by 5,764 hectares, and grapes by 10,706 hectares. By comparison, 2025 estimates show a 6,452 hectare increase in almond plantings, a 1,308 hectare increase in citrus, and an 1,899 hectare decrease in grape plantings. This equates to an 82 per cent drop in planned planting growth from the previous report and an 79 per cent drop in water use (at full maturity).

In addition, these numbers show that almonds and citrus continue to grow, but at a slower rate than in the 2022 analysis, while wine grapes have reversed their growth trend, with a decrease in planned planting area.

This slowdown in forecast plantings was driven by a range of factors, including:

- While almond prices have generally increased over the last decade, the last three years have seen some global price volatility.[16] This has affected the profitability of many almond orchards and influenced production. In turn cooling investor and irrigator interest in new developments, with many preferring to maximise returns from their existing operations. It must be noted that almond prices have been increasing in recent months, as of the writing of this report; this could influence future decisions regarding the planning of new developments.[17]
- Some of the above uncertainty has also been driven by a complex global trading environment. This includes past issues in trade relations between Australia and China (which heavily affected the wine grape sector), supply chain impacts from the COVID-19 pandemic, and current uncertainty regarding tariffs placed by the United States on Australian exports.
- California, which is responsible for most of the world's almond production, has also been managing water scarcity issues in recent years after years of dry weather. This has driven the removal of some almond orchards – 2024 was the first time in 30 years that total almond plantings had dropped in three consecutive years.[18]
- Rising inflation and interest rates – following near-decades at record lows – have put pressure on asset prices and investor returns. While Reserve Banks around the world (except in Australia) are now considering holding rates steady or slightly lowering them, rates remain relatively high. This has been factored into investment decisions and has affected the outlook for future plantings.
- As the Australian government continues its purchase of water entitlements from irrigators to meet environmental targets, there is uncertainty about future water availability for consumptive use in the southern Murray-Darling Basin.

Table 9 Future planned planting area and water use estimates for the 2025 analysis

	Total potential area (ha)	Total potential water demand (ML)

[15] Almond figure includes an additional 657 hectares of “other nuts” included in the 2022 analysis
[16] Federal Reserve of St Louis, 2025. Producer Price Index by Commodity: Farm Products: Almonds. Accessible via: https://fred.stlouisfed.org/series/WPU01190102
[17] Rabobank, 2025. Australian almond outlook – “cracking growth” beyond 2030. Accessible via: https://www.rabobank.com.au/news/media-releases/2025/australian-almond-outlook-cracking-growth-beyond-2030
[18] California Almonds, 2024. California Almond Acreage Drops Again in 2024. Accessible via: https://www.almonds.org/about-us/press-room/california-almond-acreage-drops-again-2024

New South Wales – below the choke	2,801	41,010
Victoria – below the choke	2,621	29,634
Victoria – above the choke	3,000	42,000
South Australia	-2,561	-25,610
Total connected Murray	5,861	87,034
Total Lower Murray	2,861	45,034

Table 10 showing future planned planting area and water use estimates for Scenario A and Scenario B in the 2022 analysis (note: all planned planting data was collected for the lower Murray only in the 2022 analysis)

Area	Scenario A: excluding > 1,000 ha plantings		Scenario B: all plantings	
	Total potential area (ha)	Total potential water demand (ML)	Total potential area (ha)	Total potential water demand (ML)
New South Wales – below the choke	4,568	60,867	7,368	89,067
Victoria – below the choke	3,223	41,982	9,673	132,282
South Australia	5,967	72,043	15,467	188,043
Total connected Murray	13,759	174,892	32,509	409,392

Total Lower Murray	13,759	174,892	32,509	409,392

4. What is the expected impact of Commonwealth water purchases on horticultural headroom?

4.1 Background

In response to the challenges of the Millennium Drought, Australia embarked on a major market reform process that culminated in the *Water Act 2007* and, subsequently, the *Murray-Darling Basin Plan 2012* (the Basin Plan).

The Basin Plan sets out the long-term plan to support Basin communities, industry, and the environment through the sustainable management of the Murray-Darling Basin's water resources. It aims to reset the balance between consumptive water use and water for the environment through water recovery and Sustainable Diversion Limits (SDLs). SDLs are the maximum amount of water that can be taken from the Basin while maintaining environmental sustainability. The SDLs are calculated as a long-term average across the Basin and then set for each catchment.

The original water recovery target, known as the 'Bridging the Gap' water recovery target, legislated a recovery of 2,750 GL from the consumptive pool. Amendments to the Basin Plan in 2018 reduced the target to 2,075 GL. This change was supported by two additional targets:

- 605 GL of water agreed to be recovered through supply and constraints measures, and
- an additional 450 GL/y of water to deliver environmental outcomes to downstream river channels and floodplains. This water is to be recovered through efficiency measures.

These changes were made using the Sustainable Diversion Limit Adjustment Mechanism (SDLAM), which allows SDLs to be adjusted up or down by a maximum of 5% through supply, constraint, and efficiency projects.

Table 11 Surface water recovery and SDLAM progress as of 31 December 2025

	Water recovery target	Progress towards target	Remaining recovery
Bridging the Gap	2,075 GL/y	2,062.2 GL/y	15.7 GL/y
450 GL/y for enhanced environmental outcomes	450.0 GL/y	221.2 GL/y*	228.8 GL/y
SDLAM projects	605 GL/y	291 - 415 GL/y**	190 GL - 314 GL/y

** Includes 78.2GL of over-recovered entitlement based on local targets, and 57.4 GL/y of contracted, but not yet registered water.*

*** The most recent Murray-Darling Basin assurance report estimates that, as of July 2023, there is likely to be a shortfall in the recovery targets of between 190 and 314 GL, once reconciliation is undertaken. The report indicates that the shortfall is likely to be at the higher end of this range.*

Source: AG DCCEEW, Progress on Murray-Darling Basin water recovery – AG DCCEEW (last updated 24 February 2026)

4.1.1 Restoring Our Rivers Act 2023

The *Restoring Our Rivers Act 2023* made legislative changes that enable the full delivery of the Basin Plan.

Legislative amendments extended timeframes for delivery on the Plan, including:

- The deadline to complete Sustainable Diversion Limit Adjustment Mechanism (SDLAM) supply projects has been extended to **31 December 2026.**
- Additional time for new SDLAM projects to be proposed by **June 2025.**
- The deadline for the Basin Plan to be delivered, including recovery of the 450 GL/y additional water for the environment, is **31 December 2027 (previously 30 June 2024).**

The amendments also repealed the 1,500 GL cap on Australian Government water purchases and enabled agencies to implement the recommendations made in the Water Markets Roadmap and ACCC Inquiry.

4.1.2 Water recovery in the southern Murray-Darling Basin

While the 'Bridging the Gap' targets have almost been met in the southern Murray-Darling Basin, water still needs to be recovered to meet the 450 GL/y target for enhanced environmental outcomes. In some southern Murray-Darling Basin catchments, water was over-recovered towards the 'Bridging the Gap' target and was reallocated towards the 450 GL/y target. As of November 2025, 280.2 GL remains to be recovered towards the 450 GL/y target[19].

To recover the remainder of this target, the Commonwealth Government has begun a Voluntary Water Purchase Program in 2024, through which the following activities were undertaken in the southern Murray-Darling Basin:

- **'Selected Catchments' open tender:** In July 2024, the Commonwealth Government conducted an open tender to recover 70 GL of entitlement across the Murray, Ovens, and Murrumbidgee catchments. As of November 2025, the program has concluded, with the Commonwealth finalising contractual processes for a total of 51.2 GL/y under the tender.
- **Expression of Interest (EOI) 1:** In late 2024, the first expression of interest opened for parcels 10 ML or larger[20] across the southern Murray-Darling Basin catchments, not included in the July open tender. In November 2025, Minister Watt approved the commencement of water purchases under EOI 1.
- **EOI 2:** A second EOI was opened at the same time, for large water portfolios (greater than 20 GL) in southern Murray-Darling Basin catchments. Purchases under EOI 2 commenced in February 2025 for up to 100 GL of water, with the November Ministerial approval extending this limit beyond 100 GL, resulting in a total approved water purchase program of 300 GL.

[19] Restoring Our Rivers: Delivering the 450 GL Implementation Plan – November 2025 Update
[20] ATM Documents - ATM_2024_2749: AusTender

4.2 Modelling approach and scenarios considered

To model the effect of reduced supply from future water recovery on horticultural headroom, a scenario was developed to assess the effect of water recovery upon completion of the 450 GL/y and Bridging the Gap targets.

This **hypothetical**[21] scenario does not reflect the position of the Commonwealth government, Basin State governments or WSP on which entitlements will be a target of future water recovery efforts. Only once the full recovery is complete will the exact changes in water availability by catchment will be known – this may differ from the outcomes of WSP's analysis. While details on WSP's technical approach, assumptions and data are in Appendix A, overall, under this scenario, we have assumed that:

- Any future recovery by the Commonwealth would be solely focused on the southern Murray-Darling Basin. As of 30 June 2025, 157.6 GL[22] has been recovered towards the 450 GL/y target. This means 292.4 GL (450 minus 157.6) is yet to be recovered against the estimated 2025-26 consumptive entitlement on issue.
- The Commonwealth's future water recovery will lead to purchasing new entitlements in proportions matching the Commonwealth Environmental Water Holder's (CEWH) current portfolio. Essentially, this assumes the CEWH's future portfolio of southern Murray-Darling Basin entitlements will resemble its current portfolio.[23]

The resulting total and environmental entitlement on issue under the above scenario were then combined with other water supply and usage data already discussed in this report. All other assumptions and data were held constant. This then generated a new set of horticultural headroom figures, which are detailed below with supporting figures in Figure 17.

4.3 Impact of the full 450 GL/y recovery on water availability

The difference in consumptive entitlement on issue for the southern Murray-Darling Basin between 2024-25 and full recovery of the 450 GL/y target was 363 GL.

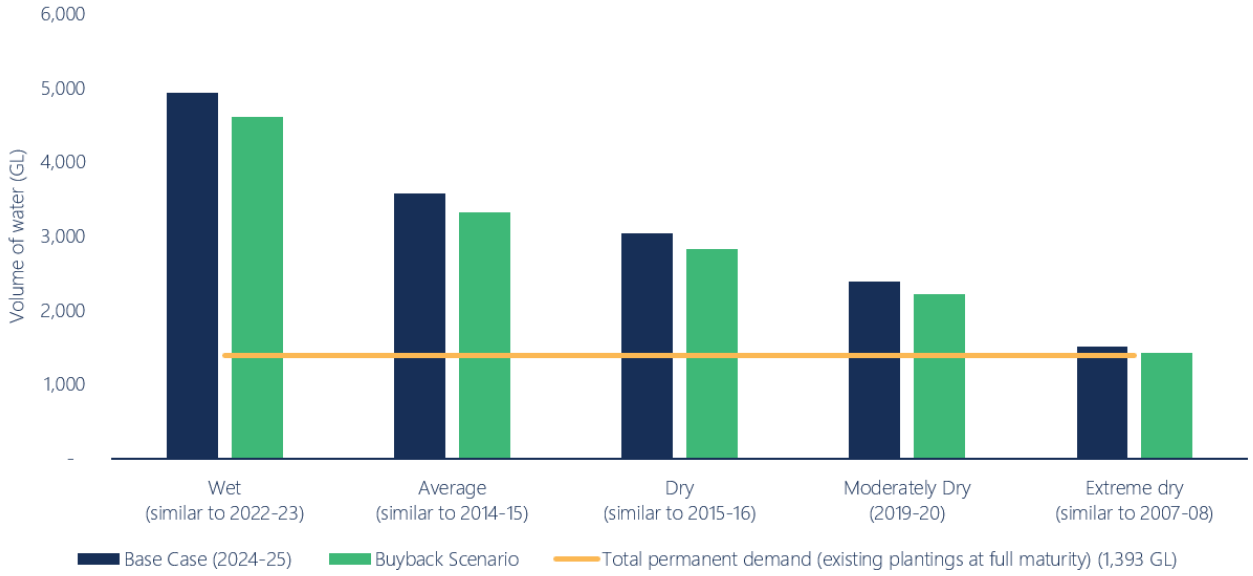
Reductions in water supply (including groundwater) across the connected Murray range from 6% to 7.2%. This increases to 7% to 9% when groundwater is excluded. Similar trends apply to the lower Murray, with an approximately 7% to 8% reduction.

Across climate scenarios, this translates to a volume reduction of 91 to 329 GL in the connected Murray and 37 to 124 GL in the lower Murray, with the higher end of the range in wet years and the lower end in extreme dry years.

In the connected Murray, this is particularly prominent under an extreme dry scenario. Even when groundwater is included, the horticultural headroom reduces from 128 GL to 37 GL. For the lower Murray, this is applicable even in a dry scenario (reducing from 128 GL to 27 GL).

[21] The hypothetical scenario has been developed for illustrative purposes for readers of this report.
[22] Australian Government, 2025. Review of the Water for the Environment Special Account (26 November 2025) – p.15. Accessible via https://www.dcceew.gov.au/water/policy/mdb/policy/wesa-review
[23] Data on current CEWH environmental holdings (as of 30 September 2025) was provided by CEWH in developing this report to support this analysis

Figure 16 Comparison of 2024-25 water supply with buyback scenario in the connected Murray under different climate scenarios. Top figure compares supply versus total existing demand at full maturity. Bottom figure shows resulting horticultural headroom



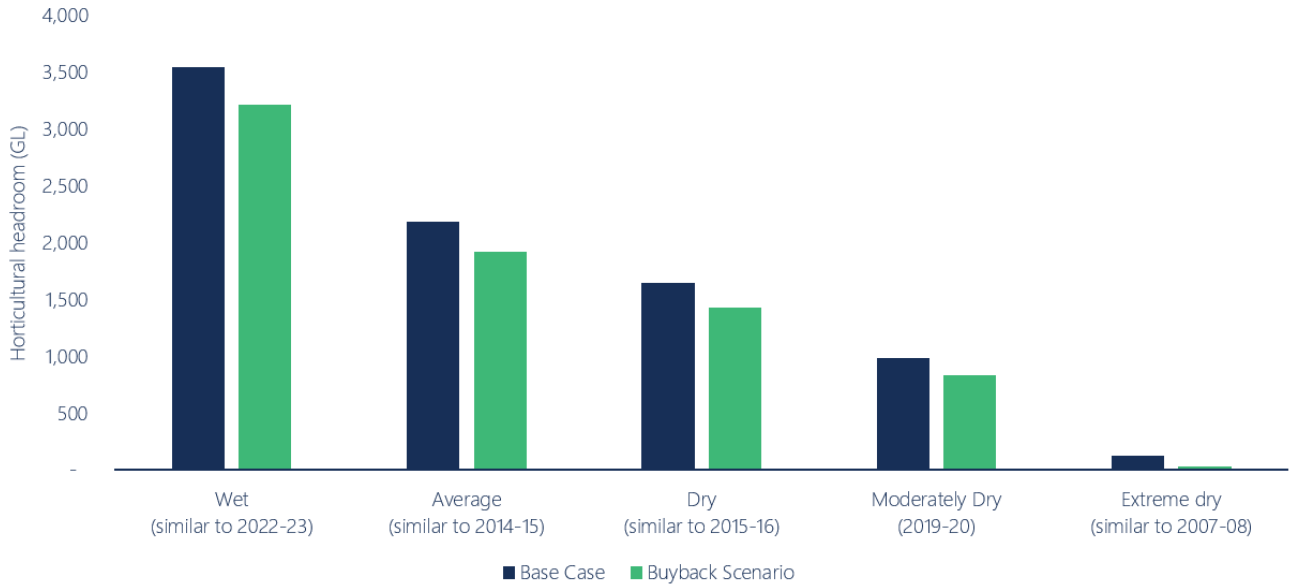
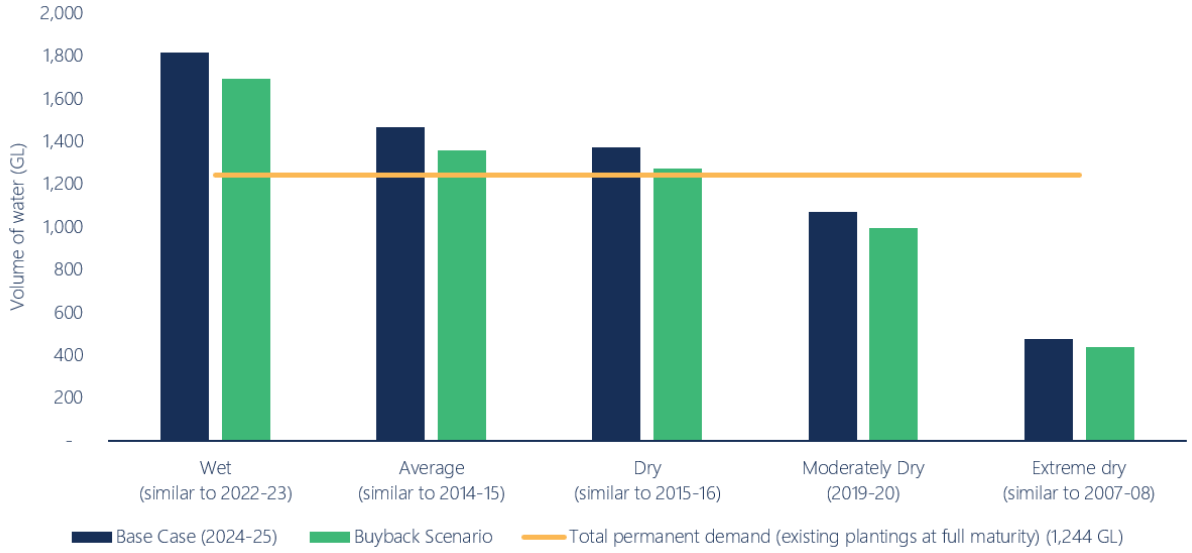
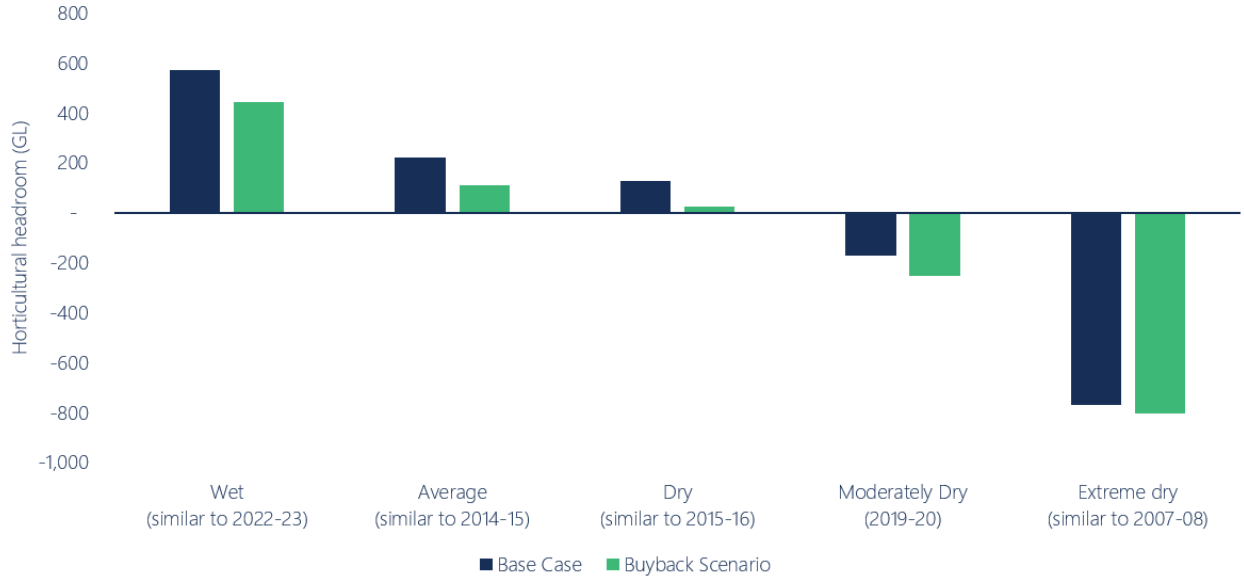


Figure 17 Comparison of 2024-25 water supply with buyback scenario in the lower Murray under different climate scenarios Top figure compares supply versus total existing demand at full maturity. Bottom figure shows resulting horticultural headroom





5. Conclusions

Access to accurate information on the supply and demand dynamics in the southern Murray-Darling Basin is critical to supporting decisions by water market participants. This report presents updated research and analysis on the growth in permanent horticulture water demand and changes in water availability in the southern Murray-Darling Basin (particularly in the connected Murray and lower Murray regions). The key conclusions are:

This is the fourth edition of this report since the original analysis was published in 2019. Over four editions, the report has been updated with the best available data, reflecting changes in methodology and addressing relevant topics of the time.

Although some data points have changed since the last report, the key messages from the 2022 report remain relevant to this analysis.

During dry periods, permanent horticultural demand in the lower Murray may exceed supply.

Consistent with previous analysis, in the lower Murray, supply is exceeded by existing permanent demand (at full maturity) under both extreme dry (-722 GL) and moderately dry (-127 GL) conditions.

This highlights the importance of this analysis for water market participants in the lower Murray. A repeat of conditions similar to the Millennium Drought would be extremely challenging for permanent horticulturalists in this region. It also highlights the continuing importance of intervalley trade and of carryover as a risk-mitigation tool for water users in the lower Murray.

New development is continuing; however, it has slowed considerably since the last report.

While demand estimates for existing permanent horticulture in the connected Murray and lower Murray are broadly consistent with the 2022 results, projections for future development vary considerably. Several factors may explain this difference. Firstly, significant development at various stages of planning and implementation occurred between 2017 and 2020; these developments are now largely mature, and their water use has stabilised. Secondly, broader economic factors, including a three-year decline in almond prices, rising interest rates, challenging trade dynamics, and uncertainty over water availability amid Commonwealth water purchases, have influenced demand. Related to this is the sharp decline in demand for winegrapes, driven by past trade tensions with China and a global oversupply. In addition, since 2019 the Victorian Government has not issued any more extraction share to new water-intensive developments in the lower Murray, formalised by a cap on total extraction share with the introduction of the “Place of Take” framework in late 2023.

The demand estimates indicate what might occur if existing plantings reach maturity, planned developments are completed, and there are no unforeseen offsetting cutbacks in other horticultural sectors. Nonetheless, there is considerable uncertainty about future outcomes, particularly regarding permanent horticultural developments at various planning stages. While there remains strong interest among developers of permanent horticultural assets in the lower Murray (below the Barmah

Narrows), uncertainty remains about which projects will proceed. Decision-making for individual enterprises will depend on a complex mix of factors, including seasonal conditions, successive dry or wet years, labour market dynamics, commodity prices, and the availability and cost of capital. For these reasons, it is essential that analyses like those in this report are regularly updated.

Further Commonwealth water purchases are likely to increase horticultural headroom pressure in dry conditions.

With horticultural headroom already under pressure in moderately dry and extreme dry conditions, any program that further reduces the consumptive pool of entitlements and, by extension, water availability is likely to be problematic for water market participants, especially in the lower Murray.

The Commonwealth Government has committed to delivering the Basin Plan in full, including delivering the additional 450 GL/y of water for the environment. This is currently pursued through several programs, including voluntary water purchases targeting entitlements in the southern Murray-Darling Basin. Since the last report, more than 140 GL/y has been recovered towards this target. In November 2025, Minister Watt announced that a further 130 GL would be recovered through voluntary water purchases in the southern Murray-Darling Basin.

Assuming the Commonwealth meets its 450 GL/y target, there could be serious implications for horticultural headroom, particularly in the lower Murray. If the remaining volume is recovered from the southern basin in proportion to the CEWH's current water portfolio, water availability in the Murray would decrease by between 6% and 7% depending on the type of year.

This reduction in water availability causes horticultural headroom to drop by 71% in the connected Murray under an extreme dry scenario. The impact on the lower Murray is much larger. The reductions turn the moderately dry scenario into a significant shortfall of 250 GL (a 45% increase), and the dry scenario becomes marginal, with horticultural headroom down to just 35 GL.

References

Aither (2019). *Water supply and demand in the southern Murray-Darling Basin*. Accessible via: https://waterregister.vic.gov.au/images/documents/Water-Supply-and-Demand-Report_Aither_FINAL.pdf

Aither (2020). *Water supply and demand in the southern Murray-Darling Basin (2020 update)*. Accessible via: https://waterregister.vic.gov.au/images/documents/Water-supply-and-horticulture-demand-in-the-sMDB-2020-Update.pdf

Aither (2022). *Water Supply and demand in the southern Murray-Darling Basin (2022 update).* Accessible via: https://waterregister.vic.gov.au/images/documents/Water-Supply-and-Demand-in-the-sMDB_2022-Update_Report.pdf

Almond Board of Australia (2025). *Almond Insights 2024/25*. Accessible via: https://viewer.joomag.com/2024-25-aba-almond-insights/0387963001763097522/p1?short=

Australian Competition and Consumer Commission (2021). *Murray-Darling Basin water markets inquiry – final report*. Available at: https://www.accc.gov.au/system/files/Murray-Darling%20Basin%20-%20water%20markets%20inquiry%20-%20Final%20report_0.pdf

California Almonds (2024). *California Almond Acreage Drops Again in 2024*. Accessible via: https://www.almonds.org/about-us/press-room/california-almond-acreage-drops-again-2024

CCW Cooperative (2025). *Riverland Transformation Pathways; a Comprehensive Industry Assessment and Strategic Framework*.

Department of Climate Change, Energy, the Environment and Water (2025). *Voluntary Water Purchase Program for the 450 GL/y (Restoring our Rivers)*. Accessible via: https://www.dcceew.gov.au/water/policy/water-recovery/government-water-purchasing/voluntary-restoring-our-rivers

Federal Reserve Bank of St Louis (2025). *Producer Price Index by Commodity: Farm Products: Almonds*. Accessible via: https://fred.stlouisfed.org/series/WPU01190102

Mallee Catchment Management Authority (2025). *Annual Report 2024-25*. Accessible via: https://malleecma.com.au/wp-content/uploads/2025/11/Mallee-CMA-Annual-Report-2024-25.pdf

Quinlivan, D. (2022). *Water market reform: final roadmap report*. Department of Climate Change, Energy the Environment and Water. Accessible via: https://www.dcceew.gov.au/sites/default/files/documents/water-market-reform-final-roadmap-report.pdf

SunRISE Mapping and Research (2025). *Irrigated crop area data for the Lower Murray-Darling 2003 to 2024.* Accessible via: https://www.mdba.gov.au/sites/default/files/publications/final-report-irrigated-crop-area-data-for-the-lower-murray-darling-2003-to-2024.pdf

Water Amendment (Restoring our Rivers) Act (2023). Available at: https://www.legislation.gov.au/C2023A00111/asmade/text

Appendices

Appendix A – Key notes on how Commonwealth water purchases were modelled to understand horticultural headroom

Key notes on the Buyback Scenario: the impact of the full 450 GL/y recovery on water availability

This scenario was developed to reflect the horticultural headroom under a hypothetical scenario where the Commonwealth has achieved its water recovery targets (both 450 GL/y and Bridging the Gap targets). As of 30 June 2025, 157.6 GL[24] has been recovered towards the 450 GL/y target. This means 292.4 GL (450 minus 157.6) is yet to be recovered. It was assumed that this outstanding volume would be recovered solely from the southern Murray-Darling Basin.

To estimate how the remaining **292.4 GL** would be recovered, it was assumed that any future recovery of entitlements would be done in a proportion that reflected CEWH's current environmental water holdings. Data on current CEWH environmental holdings (as of 30 September 2025) was provided by CEWH in developing this report to support this analysis.

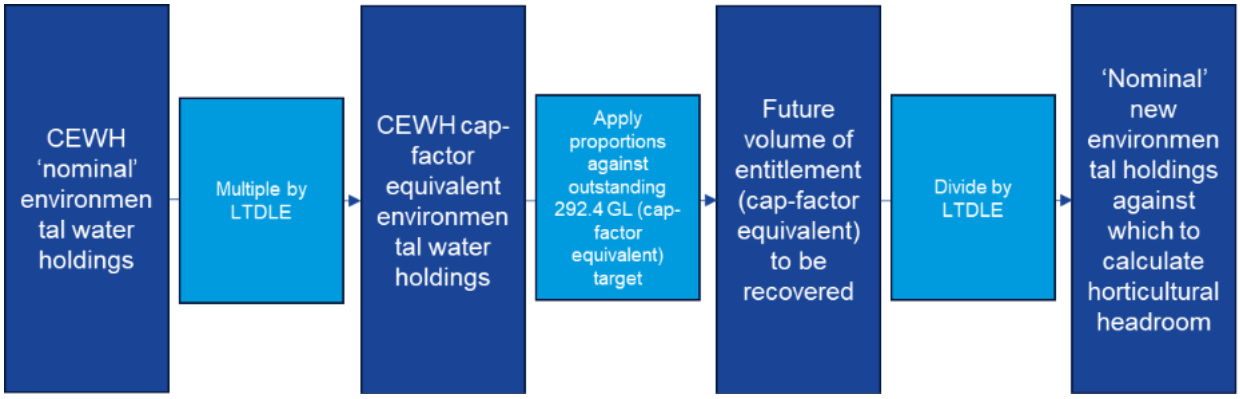
Any calculation of future entitlement purchases against environmental targets must also consider the long-term diversion limit equivalent (LTDLE) factor of the entitlement. The LTDLE (also referred to as the Cap factor) is a water accounting tool used by the Murray-Darling Basin Authority to measure how much water has been recovered for the environment as part of the Murray-Darling Basin Plan and to guide future water recovery decisions. These factors must be used when analysing the volume of water to be recovered for the environment.

$$Volume\ of\ entitlement\ \times LTDLE\ (cap\ factor) =' Cap\ factor\ equivalent' volume\ of\ entitlement$$

Figure 18 below summarises the approach used in this scenario. It must be noted that the total entitlement on issue under this scenario was assumed to be the same as 2025-26 volumes.

[24] Australian Government, 2025. Review of the Water for the Environment Special Account (26 November 2025) – p.15. Accessible via https://www.dcceew.gov.au/water/policy/mdb/policy/wesa-review

Figure 18 Summary of approach used to calculate new environmental water holdings under the Buyback Scenario.



The results of this approach are summarised in Table 12. Data on the CEWH's water holdings (as of 30 September 2025) and the entitlement LTDLE is summarised in Table 13.

Table 12 Estimate of total and environmental entitlement on issue for Buyback Scenario: the impact of the full 450 GL/y recovery on water availability.

Entitlement	Total EOI	Environmental EOI	Data source/s and assumptions
New South Wales			
NSW Murray 10 GS	1,301.275	462.837	Data sources and assumptions explained above
NSW Murray 10 HS	22.811	6.360	
NSW Murray 11 GS	372.821	144.492	
NSW Murray 11 HS	166.893	30.639	
NSW Lower Murray Groundwater	84.710	1.32	
South Australia			
SA Murray HS Class 3	608	240	Data sources and assumptions explained above
Victoria			
Vic 1A Greater Goulburn HRWS	1,015.776	389.479	Data sources and assumptions explained above
Vic 1A Greater Goulburn LRWS	445.259	60.050	
Vic 1B Boort HRWS	83.215	27.084	

Entitlement	Total EOI	Environmental EOI	Data source/s and assumptions
Vic 1B Boort LRWS	38.623	10.137	
Vic 2 Broken HRWS	17.623	0.736	
Vic 2 Broken LRWS	3.301	0.024	
Vic 3 Goulburn HRWS	20.929	7.609	
Vic 3 Goulburn LRWS	5.172	1.235	
Vic 4A Campaspe HRWS	21.869	7.209	
Vic 4A Campaspe LRWS	19.175	0.478	
Vic 4C Campaspe HRWS	1.634	0.764	
Vic 5A Loddon HRWS	21.525	4.528	
Vic 5A Loddon LRWS	8.071	0.638	
Vic 6 Murray HRWS	329.297	142.713	
Vic 6 Murray LRWS	136.450	23.986	
Vic 6B Lower Broken Creek HRWS	24.422	4.733	
Vic 6B Lower Broken Creek LRWS	12.001	0.319	
Vic 7 Murray HRWS	951.888	328.459	
Vic 7 Murray LRWS	186.643	31.067	
Vic groundwater	430.523	0.00	

Table 13 Volume of water held by the Commonwealth Environmental Water Holder and long-term diversion limit equivalent (LTDLE) factors of the entitlement. The LTDLE must be used when tracking entitlement purchases against legislated targets.

Entitlement	Volume held by Commonwealth Environmental Water Holdings (30 Sep 2025) (GL)	Long-term diversion limit equivalent factors (LTDLE)	Data source/s and assumptions
New South Wales			
NSW Murray 10 GS	331.901	0.699	CEWH holdings provided by CEWH. LTDLE factors sourced from: https://www.water.dcceew.nsw.gov.au/sites/default/files/2025-08/NSW%20updated%20factors%20for%20water%20recovery.pdf
NSW Murray 10 HS	3.010	0.873	
NSW Murray 11 GS	73.812	0.699	
NSW Murray 11 HS	21.678	0.873	
NSW Lower Murray Groundwater	1.522	1	
South Australia			
SA Murray HS Class 3	156.601	0.882	CEWH holdings provided by CEWH. LTDLE factors sourced from: https://cdn.environment.sa.gov.au/environment/docs/fact-sheet-ltdle-factors.pdf
Victoria			
Vic 1A Greater Goulburn HRWS	317.077	0.967	CEWH holdings provided by CEWH LTDLE factors sourced from: https://www.water.vic.gov.au/our-programs/murray-darling-basin/long-term-diversion-limit-equivalent-factors
Vic 1A Greater Goulburn LRWS	43.845	0.583	
Vic 1B Boort HRWS	16.693	0.967	
Vic 1B Boort LRWS	0.435	0.583	
Vic 2 Broken HRWS	0.534	0.569	

Entitlement	Volume held by Commonwealth Environmental Water Holdings (30 Sep 2025) (GL)	Long-term diversion limit equivalent factors (LTDLE)	Data source/s and assumptions
Vic 2 Broken LRWS	0.004	0.075	
Vic 3 Goulburn HRWS	3.791	0.967	
Vic 3 Goulburn LRWS	0.00	0.583	
Vic 4A Campaspe HRWS	5.962	0.949	
Vic 4A Campaspe LRWS	0.395	0.572	
Vic 4C Campaspe HRWS	0.632	0.949	
Vic 5A Loddon HRWS	3.356	0.624	
Vic 5A Loddon LRWS	0.527	0.049	
Vic 6 Murray HRWS	107.897	0.974	
Vic 6 Murray LRWS	16.878	0.543	
Vic 6B Lower Broken Creek HRWS	3.848	0.569	
Vic 6B Lower Broken Creek LRWS	0.264	0.075	
Vic 7 Murray HRWS	259.600	0.974	
Vic 7 Murray LRWS	23.494	0.543	

Entitlement	Volume held by Commonwealth Environmental Water Holdings (30 Sep 2025) (GL)	Long-term diversion limit equivalent factors (LTDLE)	Data source/s and assumptions
Vic groundwater	0.00	1	

Appendix B – Water availability data 2024-25

Approach

To estimate the volume of water available for irrigation, WSP worked with the respective state government departments to access the best available data for entitlement on issue (EOI), environmental water holdings, urban water holdings, and allocations made available to the market. All data were current for the 2024-25 water year. Data was made available as summarised in Table 15, Table 16, and Table 17.

Table 14 Summary of data accessed for estimating 2024-25 water availability

State	
New South Wales	Data were provided disaggregated, Table 16 presents the EOI for environment and Table 18 presents the EOI for consumptive / irrigation.
South Australia	Data were provided disaggregated, Table 16 presents the EOI for environment and Table 18 presents the EOI for consumptive / irrigation.
Victoria	Data were provided inclusive of environmental water holdings, and some urban water holdings. Table 16 presents the EOI for environment and Table 18 presents the EOI for consumptive / irrigation.

$$\begin{aligned} &Water\ available\ for\ irrigation \\ &= ((Total\ entitlment\ on\ issue - Environmental\ water\ holdings - Urban\ water\ holdings) \\ &* (End\ of\ season\ allocations\ under\ each\ climatic\ scenario)) \\ &+ Water\ allocations\ from\ (non) irrigation\ entitlements\ made\ available\ to\ the\ market) \end{aligned}$$

The following tables present the data used to estimate water available for irrigation in the connected Murray, as well as the data sources and any assumptions that were required.

- Table 15 presents the total entitlement on issue (EOI)
- Table 16 presents the environmental holdings
- Table 17 presents urban water holdings
- Table 18 summarises the total EOI available for irrigation
- Table 19 presents the end of season water allocation percentages
- Table 20 summarises the estimated volumes of water available for irrigation by entitlement
- Table 21 presents water allocations typically made available to the market
- Table 22 summarises the estimated volumes of water available for irrigation in the regions of interest i.e., the connected Murray and the lower Murray

The data are presented with a view to achieving as much consistency as possible across the three states to enable the analysis to be updated efficiently in the future.

Data Tables – Water availability

Total entitlement on issue

Table 15 2024-25 volumes of entitlement on issue (GL)

Entitlement	2024-25 EOI	Data source/s and assumptions
New South Wales		
NSW Murray 10 GS	1,301.236	NSW DCCEEW dashboard at: https://www.industry.nsw.gov.au/water/allocations-availability/water-accounting/share-component-dashboard. NSW DCCEEW confirmed that the EOI were provided net of urban and country town water licences and conveyance water holdings. Data include environmental water holdings and licences held by irrigation corporations. Distribution of shares across NSW 10 and 11 determined by WSP from the NSW Water Register. Result cross referenced with data sourced from NSW DCCEEW.
NSW Murray 10 HS	22.811	
NSW Murray 11 GS	372.860	
NSW Murray 11 HS	166.894	
NSW Lower Murray Groundwater	84.789	
South Australia		
SA Murray HS Class 3	607.8	Provided by DEW. Data includes held environmental water. Net of urban and country town water holdings. When South Australia receives an entitlement of greater than 1,500 GL this is the distribution to consumptive / irrigation holdings and environmental holdings.
Victoria		
Vic 1A Greater Goulburn HRWS	1,015.776	Provided by DEECA. This only refers to water shares as there are also smaller volumes of take and use licences, supply by agreements and bulk entitlements. This data includes Environmental Water Holdings and Water Corporation Holdings. .
Vic 1A Greater Goulburn LRWS	445.259	
Vic 1B Boort HRWS	83.215	
Vic 1B Boort LRWS	38.623	
Vic 2 Broken HRWS	17.623	
Vic 2 Broken LRWS	3.301	
Vic 3 Goulburn HRWS	20.929	
Vic 3 Goulburn LRWS	5.172	
Vic 4A Campaspe HRWS	21.869	
Vic 4A Campaspe LRWS	19.175	
Vic 4C Campaspe HRWS	1.634	

Entitlement	2024-25 EOI	Data source/s and assumptions
Vic 5A Loddon HRWS	21.525	
Vic 5A Loddon LRWS	8.071	
Vic 6 Murray HRWS	329.297	
Vic 6 Murray LRWS	136.450	
Vic 6B Lower Broken Creek HRWS	24.422	
Vic 6B Lower Broken Creek LRWS	12.001	
Vic 7 Murray HRWS	951.888	
Vic 7 Murray LRWS	186.643	
Vic groundwater	430.523	

Environmental water holdings

Table 16 2024-25 environmental water holdings (GL)

Entitlement	Total holdings	Data source/s and assumptions
New South Wales		
NSW Murray 10 GS	378.791	NSW DCCEEW dashboard at: https://www.industry.nsw.gov.au/water/environmental-water-hub/public-register/dashboard/held NSW Lower Murray Groundwater environmental data provided separately by NSW DCCEEW via email on 1 September 2025 Only total environmental holdings were provided. Distribution of shares across NSW 10 and 11 determined by WSP from the public register of environmental water holdings in NSW Murray. Result cross referenced with data sourced from NSW DCCEEW.
NSW Murray 10 HS	4.699	
NSW Murray 11 GS	124.526	
NSW Murray 11 HS	21.422	
NSW Lower Murray Groundwater	1.320	
South Australia		
SA Murray HS Class 3	207.71	Provided by DEW. This water is held by the Commonwealth Environmental Water Office and the South Australian Government for environmental purposes. When South Australia receives an entitlement of greater than 1,526 GL this volume is allocated to the environment. This includes an additional 1 GL of water allocated for consumptive purposes which is to be used for the environment each year as advised by DEW, and has been deducted from the EOI for irrigation.

Entitlement	Total holdings	Data source/s and assumptions
		This does not include an additional 12 GL required for the environment when SA Entitlement on Issue reaches 1,600 GL under the Adelaide Desalination Plant Agreement with the Commonwealth and an associated Section 6 directive under the Public Corporations Act. This 12 GL is acquired from the remaining 401 GL allocated for consumptive purposes.
Victoria		
Vic 1A Greater Goulburn HRWS	321.031	Provided by DEECA. This only refers to water shares as there are also smaller volumes of take and use licences, supply by agreements and bulk entitlements
Vic 1A Greater Goulburn LRWS	50.852	
Vic 1B Boort HRWS	23.594	
Vic 1B Boort LRWS	10.046	
Vic 2 Broken HRWS	0.624	
Vic 2 Broken LRWS	0.023	
Vic 3 Goulburn HRWS	6.816	
Vic 3 Goulburn LRWS	1.235	
Vic 4A Campaspe HRWS	5.962	
Vic 4A Campaspe LRWS	0.395	
Vic 4C Campaspe HRWS	0.632	
Vic 5A Loddon HRWS	3.826	
Vic 5A Loddon LRWS	0.527	
Vic 6 Murray HRWS	120.088	
Vic 6 Murray LRWS	20.458	
Vic 6B Lower Broken Creek HRWS	3.929	
Vic 6B Lower Broken Creek LRWS	0.264	
Vic 7 Murray HRWS	274.193	
Vic 7 Murray LRWS	26.156	
Vic groundwater	0.000	

Urban water holdings

Table 17 2024-25 volumes of urban water holdings within the total EOI.

Entitlement	Urban water holdings	Data source/s and assumptions
New South Wales		
NSW Murray 10 GS	0.000	NSW EOI data presented above are net of any urban entitlements, and net of conveyance water, including MIL conveyance as confirmed by NSW DCCEEW. Assumption: all urban water requirements are met by town water licences.
NSW Murray 10 HS	0.000	
NSW Murray 11 GS	0.000	
NSW Murray 11 HS	0.000	
NSW Lower Murray Groundwater	0.000	
South Australia		
SA Murray HS Class 3	0.000	SA EOI data presented above are net of any urban entitlements. Conveyance water does not apply in South Australia.
Victoria		
Vic 1A Greater Goulburn HRWS	31.541	Provided by DEECA. This only refers to water shares as there are also smaller volumes of take and use licences, supply by agreements and bulk entitlements Conveyance water does not apply in Victoria. Vic total EOI data presented above include some urban water holdings.
Vic 1A Greater Goulburn LRWS	4.676	
Vic 1B Boort HRWS	5.603	
Vic 1B Boort LRWS	0.000	
Vic 2 Broken HRWS	1.144	
Vic 2 Broken LRWS	0.065	
Vic 3 Goulburn HRWS	0.000	
Vic 3 Goulburn LRWS	0.000	
Vic 4A Campaspe HRWS	2.692	
Vic 4A Campaspe LRWS	12.088	
Vic 4C Campaspe HRWS	0.000	
Vic 5A Loddon HRWS	0.036	
Vic 5A Loddon LRWS	0.000	
Vic 6 Murray HRWS	1.746	
Vic 6 Murray LRWS	0.231	

Entitlement	Urban water holdings	Data source/s and assumptions
Vic 6B Lower Broken Creek HRWS	0.149	
Vic 6B Lower Broken Creek LRWS	0.002	
Vic 7 Murray HRWS	7.334	
Vic 7 Murray LRWS	1.356	
Vic groundwater	7.476	

Entitlement on issue available for irrigation

Table 18 Total EOI volumes available for irrigation (GL)

Entitlement	EOI for irrigation	Data source/s and assumptions
New South Wales		
NSW Murray 10 GS	922.4450	Formula: - NSW Water Available for Irrigation = Total EOI - Environmental Holdings
NSW Murray 10 HS	18.1120	
NSW Murray 11 GS	248.3340	
NSW Murray 11 HS	145.4720	
NSW Lower Murray Groundwater	83.4690	
South Australia		
SA Murray HS Class 3	400.0900	Formula: - SA Water Available for Irrigation = Total EOI - Environmental Holdings
Victoria		
Vic 1A Greater Goulburn HRWS	663.2045	Formula: - VIC Water Available for Irrigation = Total EOI - Environmental Holdings - Urban holdings
Vic 1A Greater Goulburn LRWS	389.7314	
Vic 1B Boort HRWS	54.0179	
Vic 1B Boort LRWS	28.5766	
Vic 2 Broken HRWS	15.8550	
Vic 2 Broken LRWS	3.2126	
Vic 3 Goulburn HRWS	14.1126	
Vic 3 Goulburn LRWS	3.9374	
Vic 4A Campaspe HRWS	13.2147	
Vic 4A Campaspe LRWS	6.6913	
Vic 4C Campaspe HRWS	1.0015	
Vic 5A Loddon HRWS	17.6630	
Vic 5A Loddon LRWS	7.5438	
Vic 6 Murray HRWS	207.4629	
Vic 6 Murray LRWS	115.7613	
Vic 6B Lower Broken Creek HRWS	20.3442	

Entitlement	EOI for irrigation	Data source/s and assumptions
Vic 6B Lower Broken Creek LRWS	11.7344	
Vic 7 Murray HRWS	670.3598	
Vic 7 Murray LRWS	159.1314	
Vic groundwater	423.0470	

Climate scenarios

Table 19 Climate scenarios: End of season water allocation percentages

Entitlement	Wet (similar to 2022-23)	Average (similar to 2014-15)	Dry (similar to 2015-16)	Moderately dry (similar to 2019-20)	Extreme dry (similar to 2007-08)	Notes and assumptions
New South Wales						Wet scenario updated from 2011-12 in 2022 to 2022-23 in 2025. Applies to SA and Vic also
NSW 10 Murray GS	110%	61%	23%	3%	0%	
NSW 10 Murray HS	100%	97%	97%	97%	25%	
NSW 11 Murray GS	110%	61%	23%	3%	0%	
NSW 11 Murray HS	100%	97%	97%	97%	25%	
NSW Lower Murray groundwater	100%	100%	100%	100%	100%	
South Australia						No groundwater in SA Murray
SA Murray HS Class 3	100%	100%	100%	100%	36%[25]	
Victoria						It must be noted that environmental water holder and water corporation trading activity likely depends on many factors, not just the climate of the year. For example, the climate of previous years can influence the volume of water available for trading. The
Vic 1A Greater Goulburn HRWS	100%	100%	90%	80%	57%	
Vic 1A Greater Goulburn LRWS	100%	0%	0%	0%	0%	
Vic 1B Boort HRWS	100%	100%	90%	80%	57%	
Vic 1B Boort LRWS	100%	0%	0%	0%	0%	
Vic 2 Broken HRWS	100%	100%	26%	2%	71%	
Vic 2 Broken LRWS	100%	100%	0%	0%	0%	
Vic 3 Goulburn HRWS	100%	100%	90%	80%	57%	
Vic 3 Goulburn LRWS	100%	0%	0%	0%	0%	

[25] SA DEW advised that while allocations in this year were 32%, following the consolidation of multiple entitlement types, effective allocations to Class 3 in 2007-08 would have been 36%.

Entitlement	Wet (similar to 2022-23)	Average (similar to 2014-15)	Dry (similar to 2015-16)	Moderately dry (similar to 2019-20)	Extreme dry (similar to 2007-08)	Notes and assumptions
Vic 4A Campaspe HRWS	100%	100%	66%	80%	100%	health of various water sites can also influence the priorities of the environmental water holder and their decision to trade (or not to). Broader business trading strategies and approaches can also change over time and influence trading decisions
Vic 4A Campaspe LRWS	100%	0%	0%	0%	0%	
Vic 4C Campaspe HRWS	100%	100%	66%	180%	100%	
Vic 5A Loddon HRWS	100%	100%	84%	80%	3%	
Vic 5A Loddon LRWS	100%	0%	0%	0%	0%	
Vic 6 Murray HRWS	100%	100%	100%	66%	43%	
Vic 6 Murray LRWS	100%	0%	0%	0%	0%	
Vic 6B Lower Broken Creek HRWS	100%	100%	100%	66%	43%	
Vic 6B Lower Broken Creek LRWS	100%	0%	0%	0%	0%	
Vic 7 Murray HRWS	100%	100%	100%	66%	43%	
Vic 7 Murray LRWS	100%	0%	0%	0%	0%	
Vic groundwater	100%	100%	100%	100%	100%	

Summary 1: Water availability by entitlement

Table 20 Estimate of total water available for irrigation by entitlement (GL)

Entitlement	Wet (similar to 2022-23)	Average (similar to 2014-15)	Dry (similar to 2015-16)	Moderately dry (similar to 2019-20)	Extreme dry (similar to 2007-08)
New South Wales					
NSW 10 Murray GS	1,015	563	212	28	-
NSW 10 Murray HS	18	18	18	18	5
NSW 11 Murray GS	273	151	57	7	-
NSW 11 Murray HS	145	141	141	141	36
NSW Lower Murray groundwater	83	83	83	83	83
South Australia					
SA Murray HS Class 3	400	400	400	400	144
Victoria					
Vic 1A Greater Goulburn HRWS	663	663	597	531	378
Vic 1A Greater Goulburn LRWS	390	-	-	-	-
Vic 1B Boort HRWS	54	54	49	43	31
Vic 1B Boort LRWS	29	-	-	-	-
Vic 2 Broken HRWS	16	16	4	0	11
Vic 2 Broken LRWS	3	3	-	-	-
Vic 3 Goulburn HRWS	14	14	13	11	8
Vic 3 Goulburn LRWS	4	-	-	-	-
Vic 4A Campaspe HRWS	13	13	9	11	13

Entitlement	Wet (similar to 2022-23)	Average (similar to 2014-15)	Dry (similar to 2015-16)	Moderately dry (similar to 2019-20)	Extreme dry (similar to 2007-08)
Vic 4A Campaspe LRWS	7	-	-	-	-
Vic 4C Campaspe HRWS	1	1	1	1	1
Vic 5A Loddon HRWS	18	18	15	14	1
Vic 5A Loddon LRWS	8	-	-	-	-
Vic 6 Murray HRWS	207	207	207	137	89
Vic 6 Murray LRWS	116	-	-	-	-
Vic 6B Lower Broken Creek HRWS	20	20	20	13	9
Vic 6B Lower Broken Creek LRWS	12	-	-	-	-
Vic 7 Murray HRWS	670	670	670	442	288
Vic 7 Murray LRWS	159	-	-	-	-
Vic groundwater	423	423	423	423	423
Sub-total	6,798	4,824	4,049	3,064	2,288

Other water allocations made available to the market

Table 21 Other water holdings that are made available to the market (GL)

Entitlement type	Wet (similar to 2022-23)	Average (similar to 2014-15)	Dry (similar to 2015-16)	Moderately dry (similar to 2019-20)	Extreme dry (similar to 2007-08)	Notes and assumptions
New South Wales						NSW DCCEEW confirmed no other water is regularly made available to the market
NSW above Barmah Narrows	0	0	0	0	0	
NSW below Barmah Narrows	0	0	0	0	0	
South Australia						Provided by DEW
SA Murray HS Class 3	8	8	8	0	0	
Victoria						Provided by DEECA.
Vic above Barmah Narrows and Goulburn	48	35	35	13	0	Environmental water holder and water corporation trading activity likely depends on many factors, not just the climate of the year. For example, the climate of previous years can influence the volume of water available for trading. The health of various water sites can also influence the priorities of the environmental water holder and their decision to
Vic below Barmah Narrows	127	75	75	68	0	

Entitlement type	Wet (similar to 2022-23)	Average (similar to 2014-15)	Dry (similar to 2015-16)	Moderately dry (similar to 2019-20)	Extreme dry (similar to 2007-08)	Notes and assumptions
						trade (or not to). Broader business trading strategies and approaches can also change over time and influence trading decisions.

Summary 2: Water availability in the connected Murray and lower Murray

Table 22 Estimate of total water available for irrigation by region (GL)

Entitlement type	Wet (similar to 2022-23)	Average (similar to 2014-15)	Dry (similar to 2015-16)	Moderately dry (similar to 2019-20)	Extreme dry (similar to 2007-08)
Connected Murray					
Connected Murray	4,945	3,578	3,038	2,385	1,521
Connected Murray (no groundwater)	4,438	3,072	2,531	1,879	1,014
Lower Murray					
Lower Murray (surface water only)	1,815	1,467	1,372	1,072	477

Note: We assume there is no groundwater available in the lower Murray region from the Lower Murray Groundwater Aquifer. In the 2020 analysis, a map of extraction sites was accessed. It identified that for practical purposes, almost all extraction sites were within the Murray Irrigation Limited district. This suggests that most groundwater users who also have access to surface water would access water from above the Barmah Narrows. The same assumption has been applied in the 2022 analysis.

Note: In addition, there are known groundwater quality issues in the Lower Murray Groundwater Aquifer that are actively managed, including by salt interception schemes to lower the water table and prevent it connecting to the river (personal communication, NSW DPIE, 17 January 2020). Therefore, for the lower Murray region, a surface water only water availability scenario is presented as we assume no groundwater is available in this region.

Appendix C – Water usage data 2024-25

Approach

To estimate the water demand from permanent horticulture, WSP worked with the respective state departments to access the best available data for actual water usage in 2024-25 (where available).

All data were current for the 2024-25 water year. Data were made available from the respective state departments as described in Table 23.

Table 23 Summary of data accessed for estimating 2024-25 water usage

State	Data source/s and assumptions
New South Wales	Data provided by NSW DCCEEW
South Australia	Data provided by SA DEW
Victoria	Data provided by Agriculture Victoria in consultation with Vic DEECA The data were provided soon after the end of the 2024-25 water year, before full analysis and validation could be completed. It is likely that these estimates will differ slightly from the fully validated data presented in the 2024-25 Victorian Water Accounts, which are yet to be completed.

The following tables present the data used to estimate water usage by permanent horticulture in the connected Murray and lower Murray:

- Table 24 presents the estimate of water usage in New South Wales
- Table 25 presents the estimate of water usage in Victoria
- Table 26 presents the estimate of water usage in South Australia

The estimates for water usage by permanent horticulture in the connected Murray and lower Murray are summarised in Table 27 and Table 28 respectively.

Data Tables – Water usage

New South Wales water usage

Table 24 2024-25 New South Wales water usage (GL)

Region	Total usage	Estimate of usage by permanent horticulture	Data source/s and assumptions
NSW Murray above Choke (surface water)			
Total above Picnic Point	72.939	20.675	Provided by NSW DCCEEW. Assumes no permanent hort. Excludes usage on environmental licences. For the purposes of this analysis, total MIL usage is included in the 73 GL from NSW DCCEEW. Estimate for permanent horticulture obtained from p.20 of MIL Compliance Report (2024-25), released 31 Oct 2025 - summed same crops as was done in original analysis. Only the estimate for usage by permanent horticulture applies to this analysis.
NSW Murray below Choke (surface water)			
Total below Picnic Point	333.014	138.040	Provided by NSW DCCEEW. Excludes usage on environmental licences. WSP assumes all usage downstream of Swan Hill is permanent horticulture = 138 GL. The 138 GL likely includes some annual horticulture in NSW Sunraysia and likely excludes some permanent horticulture between Picnic Point and Swan Hill.
Western Murray Irrigation Ltd.	0.000	0.000	Not applicable to this analysis. For the purposes of this assessment, WMI usage is assumed to be included in the 333 GL from NSW DCCEEW.
Groundwater			
NSW Lower Murray Groundwater	58.000	0.000	NSW Usage dashboard (Aquifer licence category). https://www.water.dcceew.nsw.gov.au/our-work/allocations-and-availability/water-accounting/usage-dashboard No usage by permanent horticulture in NSW Sunraysia.
Total - NSW	**463.953**	**158.715**	

Victorian water usage

Table 25 2024-25 Victorian water usage (GL)

Region	Total usage	Estimate of usage by permanent horticulture	Data source/s and assumptions
Goulburn-Murray Irrigation District (excluding Torrumbarry)	1089.057	86.433	Provided by Goulburn-Murray Water, Lower Murray Water and Agriculture Victoria.
Tresco	Included in Torrumbarry	Included in Torrumbarry	
Woorinen	Included in Torrumbarry	Included in Torrumbarry	
Nyah	Included in Torrumbarry	Included in Torrumbarry	
Torrumbarry	287.346	17.774	
Victorian lower Murray	605.951	528.515	
VIC Murray above Choke and in Goulburn	1089.057	86.433	Includes Goulburn-Murray Irrigation District as reported above.
VIC Murray below Choke	893.297	546.289	Includes Torrumbarry and Victorian lower Murray as reported above.
Total – Victoria	**1,982**	**633**	

South Australian water usage

Table 26 2024-25 South Australian water usage (GL)

Region	Total usage	Estimate of usage by permanent horticulture	Data source/s and assumptions
SA Murray HS Class 3	527	448	Provided by DEW Water usage data by use / extraction point are not available. Assumption: DEW estimates 85% of the annual actual take (Irrigation for SA Murray + Basin Rights) is attributed to permanent horticulture. While 75% of the total area is under permanent horticulture, further analysis is required to identify the specific crops within each.

Region	Total usage	Estimate of usage by permanent horticulture	Data source/s and assumptions
Total – South Australia	**527**	**448**	

Summary: Water usage by permanent horticulture in the connected Murray

Table 27 2024-25 Water usage in the connected Murray (GL)

Region	Estimate of usage by permanent horticulture	Notes and assumptions
New South Wales	159	Calculation Total permanent demand (at full maturity) = Updated 2024-25 baseline estimate (GL) + Additional water use from known recent permanent developments (demand at full maturity) + Planned / in discussion permanent developments (demand at full maturity)
Victoria	633	
South Australia	448	
Total – connected Murray	**1,239**	

Summary: Water usage by permanent horticulture in the lower Murray

Table 28 2024-25 Water usage in the lower Murray (GL)

Region	Estimate of usage by permanent horticulture	Notes and assumptions
New South Wales	138	Calculation Total permanent demand (at full maturity) = Updated 2024-25 baseline estimate (GL) + Additional water use from known recent permanent developments (demand at full maturity) + Planned / in discussion permanent developments (demand at full maturity)
Victoria	546	
South Australia	448	
Total – lower Murray	**1,132**	

Appendix D – Data for estimating additional water demand from existing plantings and future plantings

Approach

To estimate the additional volume of water required to meet existing permanent horticultural demand (i.e., the gap between the 2024-25 water usage estimates and the maximum water usage when existing plantings are fully mature) and the potential additional volume of water required by future permanent plantings, WSP accessed estimates from the sources shown in Table 29.

Additional water usage from existing plantings was calculated using the following formula:

$$\begin{aligned} &Additional\ Water\ Usage\ from\ Existing\ Plantings \\ &= (Water\ usage\ of\ existing\ almond\ plantings\ at\ full\ maturity \\ &- Current\ water\ usage\ of\ existing\ almond\ plantings) \end{aligned}$$

The following sections present the data and assumptions used to estimate the water demand profile for existing plantings across New South Wales, Victoria and South Australia, respectively.

These tables are presented to provide full transparency regarding the approach used to generate the water demand estimate at full maturity. There is likely to be considerable inter-annual variation and water demand is unlikely to increase in a linear fashion, so these data should not be used as estimates for water demand from existing permanent horticulture plantings in individual water years.

Additional water usage from likely future plantings was estimated by industry experts in each of the states. This included permanent horticultural developments are at various stages of approvals for which there was a reasonable level of confidence that they would go ahead in the next three to five years. A state-based summary of the maximum water usage of these developments is provided in Table 55 and Table 56.

Further detail about the approach and relevant assumptions is provided below.

Estimating additional water usage from existing permanent plantings

To derive an estimate of additional water usage from existing permanent horticultural plantings, crops included in the analysis were almonds, citrus and grapes (including table, wine and dried varieties). The assumptions are based on the changes in the area under permanent horticulture as reported in the latest SunRISE Mapping and Research crop area reports commissioned by the Murray-Darling Basin Authority (SunRISE 2025).

At the time of this analysis (2025), the most recent year for which crop area data are available was 2024. These data show that almonds and citrus expanded in area while grapes (including table, wine and dried varieties) contracted in area.

Table 29, Table 30, and Table 31 present summaries of the crop area data used to estimate the maturity profile of existing almond plantings across New South Wales, Victoria and South Australia.

Only crops that have materially increased in area planted are considered as contributors to the baseline estimate in 2024-25. It is these newer crops that are yet to reach full maturity that are included in the water usage at full maturity estimate.

Table 29 Summary of crop area data used to estimate maturity profile of existing plantings, New South Wales

Crop	Change in land area 2015-2018	Change in land area 2018-2021	Change in land area 2021-2024
Almonds	Total = 1745 ha Per year= 582 ha	Total = 3340 ha Per year = 1113 ha	Total = 620 ha Per year = 207 ha
Citrus	Total = -520 ha Per year = -173 ha	Total = 575 ha Per year = 192 ha	Total = 1080 ha Per year = 360 ha
Grapes (table, wine & dried)	Total = -75 ha Per year = -25 ha	Total = 860 ha Per year = 287 ha	Total = 145 ha Per year = 48 ha

Note: All crop areas presented above are sourced from SunRISE Mapping and Research. (June 2025) Irrigated crop area data for the Lower Murray-Darling 2003 – 2024 including analysis of the Edward/Kolety and Wakool river system. The NSW crop areas were derived from the Lower Murray-Darling NSW estimate presented in the report, minus the crop areas recorded for the Lower Darling Menindee to Wentworth weir pool (River Reach 13 A) and Lower Murrumbidgee. Calculated per annum change assumes a constant rate of change. The Grapes (table, wine and dried) category includes grapevine – dried, grapevine – table and grapevine – wine categories.

Table 30 Summary of crop area data used to estimate maturity profile of existing plantings, Victoria

Crop	Change in land area 2015-2018	Change in land area 2018-21	Change in land area 2021-2024
Almonds	Total = 3,875 ha Per year = 1,292 ha	Total = 1,930 ha Per year = 643 ha	Total = -40 ha Per year = -13 ha
Citrus	Total = 370 ha Per year = 123 ha	Total = 565 ha Per year = 188 ha	Total = 930 ha Per year = 310 ha
Grapes (table, wine & dried)	Total = -630 ha Per year = -210 ha	Total = 980 ha Per year = 327 ha	Total = -595 ha Per year = -198 ha

Note: All crop areas presented above are sourced from SunRISE Mapping and Research. (June 2025) Irrigated crop area data for the Lower Murray-Darling 2003 – 2024 including analysis of the Edward/Kolety and Wakool river system. The Grapes (table, wine and dried) category includes grapevine – dried, grapevine – table and grapevine – wine categories.

Note: Calculated per annum change assumes a constant rate of change.

Table 31 Summary of crop area data used to estimate maturity profile of existing plantings, South Australia

Crop	Change in land area 2015-2018	Change in land area 2018-21	Change in land area 2021-2024
Almonds	Total = 3,710 ha Per year = 1,237 ha	Total = 2,620 ha Per year = 873 ha	Total = 3,020 ha Per year = 1,007 ha
Citrus	Total = 150 ha Per year = 50 ha	Total = 160 ha Per year = 53 ha	Total = 285 ha Per year = 95 ha
Grapes (table, wine & dried)	Total = 25 ha Per year = 8 ha	Total = 355 ha Per year = 118 ha	Total = -1,935 ha Per year = -645 ha

Note: All crop areas presented above are sourced from SunRISE Mapping and Research. (June 2025) Irrigated crop area data for the Lower Murray-Darling 2003 – 2024 including analysis of the Edward/Kolety and Wakool river system. The Grapes (table, wine and dried) category includes grapevine – dried, grapevine – table and grapevine – wine categories.

Note: Calculated per annum change assumes a constant rate of change.

Existing plantings: Water usage assumptions for plantings yet to reach full maturity

Water application rates for irrigated agriculture depend on many factors, including region, seasonal climate, weed control of younger plantings, mounds or flat orchards, soil types, crop load, and tree density.

Our estimates for the potential maximum water usage of existing almond plantings at full maturity are based on water application rates provided by the Almond Board of Australia during 2020 of the analysis. Our estimates for citrus and grape crops are based on numbers provided from industry professionals. Estimates are provided in Table 32.

Table 32 Assumed water application rates for plantings until full maturity, by crop type (ML per hectare)

	Almonds[1]	Citrus[2, 3]	Grapes[2, 3, 4]
Year 1	2.5	4.0	3.0
Year 2	5.0	5.5	6.0
Year 3	9.5	9.0	10.0
Year 4	10.5	10.0	
Year 5	11.5	11.0	
Year 6	12.5	11.5	
Year 7	13.0	12.0	
Year 8	14.0		

1. Almond Board of Australia during the 2020 analysis. The water usage estimates for maturing almond trees presented in this report are based on the upper bounds of the water application rates presented below.

2. Discussions with industry professionals for the 2022 analysis. Grapes category includes grapevine – dried, grapevine – table and grapevine – wine categories.

3. The number of years to maturity is assumed to be 8 years for almonds, 7 years for citrus and 3 years for grapes (table, dried and wine).

4. Water application rates vary across dried, wine and table grapes. However, due to data limitations an average water application rate was applied to all existing grape crops and potential grape developments.

Existing almond plantings: Water usage assumptions for almond plantings yet to reach full maturity

Water application rates for maturing almond trees depend on many factors, including region, seasonal climate, weed control of younger plantings, mounds or flat orchards, soil types, crop load, and tree density.

Our estimates for the potential maximum water usage of existing almond plantings at full maturity are based on water application rates provided by the Almond Board of Australia.

These application rates should be considered a guide and are based on almond orchards using drip irrigation with a tree density of approximately 325 trees/ha. It should be noted that the higher level of the range is used in a dry year such as 2019.

- Year 1 = 1 – 2.5 /ha
- Year 2 = 3.5 – 5 /ha
- Year 3 = 7.5 – 9.5 /ha
- Year 4 = 9 – 10.5 /ha
- Year 5 = 10 – 11.5 /ha
- Year 6 = 11 – 12.5 /ha
- Year 7 = 11.5 – 13 /ha
- Year 8 = 12 – 14 /ha (Maximum water use)

Assumptions for reductions in water usage from reductions in planted area

The increase in planting area estimates water usage using the maturation profile presented above. However, WSP has assumed that any reductions in planted area involved the removal of trees at full maturity. As such, a net reduction in planted area assumes a reduction in water use equivalent to the reduced area multiplied by the full maturity water use of the relevant crop.

Data Tables – Additional water usage existing plantings in New South Wales

Estimated additional water usage of existing almond plantings per year – New South Wales, lower Murray

Table 33 Estimated area of almond plantings 2017-18 to 2024-25 – New South Wales (ha)

Age of trees	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32
Year 1	582	1,113	1,113	1,113	207	207	207	0							
Year 2		582	1,113	1,113	1,113	207	207	207	0						
Year 3			582	1,113	1,113	1,113	207	207	207	0					
Year 4				582	1,113	1,113	1,113	207	207	207	0				
Year 5					582	1,113	1,113	1,113	207	207	207	0			
Year 6						582	1,113	1,113	1,113	207	207	207	0		
Year 7							582	1,113	1,113	1,113	207	207	207	0	
Year 8								582	1,113	1,113	1,113	207	207	207	0
Year 9									582	1,113	1,113	1,113	207	207	207
Year 10										582	1,113	1,113	1,113	207	207
Year 11											582	1,113	1,113	1,113	207
Year 12												582	1,113	1,113	1,113
Year 13													582	1,113	1,113
Year 14														582	1,113
Year 15															582

Note: Estimated area of almond plantings from 2012-2024 sourced from SunRISE Mapping and Research (June 2025). The NSW crop areas were derived from the Lower Murray-Darling NSW estimate presented in report, minus the crop areas recorded for the Lower Darling Menindee to Wentworth weir pool (River Reach 13 A) and Lower Murrumbidgee.

Table 34 Estimated water usage of existing almond plantings, not yet fully mature – New South Wales (ML)

Age of trees	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32
Year 1	1,454	2,783	2,783	2,783	517	517	517								
Year 2		2,908	5,567	5,567	5,567	1,033	1,033	1,033							
Year 3			5,526	10,577	10,577	10,577	1,963	1,963	1,963						
Year 4				6,108	11,690	11,690	11,690	2,170	2,170	2,170					
Year 5					6,689	12,803	12,803	12,803	2,377	2,377	2,377				
Year 6						7,271	13,917	13,917	13,917	2,583	2,583	2,583			
Year 7							7,562	14,473	14,473	14,473	2,687	2,687	2,687		
Year 8								8,143	15,587	15,587	15,587	2,893	2,893	2,893	
Year 9									8,143	15,587	15,587	15,587	2,893	2,893	2,893
Year 10										8,143	15,587	15,587	15,587	2,893	2,893
Year 11											8,143	15,587	15,587	15,587	2,893
Year 12												8,143	15,587	15,587	15,587
Year 13													8,143	15,587	15,587
Year 14														8,143	15,587
Year 15															8,143
Estimate of current water usage (2024-25)								**54,503**							
Estimate of water usage at maturity (2031-32)															**63,583**
Estimate of additional water usage of existing plantings															**9,080**

Note: Water application rates for almond trees from year 1 to year 8 were sourced from the Almond Board of Australia during the 2020analysis (see above).

Estimated additional water usage of existing citrus plantings per year – New South Wales, lower Murray

Table 35 Estimated area of citrus plantings 2017-18 to 2024-25 – New South Wales (ha)

Age of trees	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32
Year 1	-173	192	192	192	360	360	360	500							
Year 2		-173	192	192	192	360	360	360	500						
Year 3			-173	192	192	192	360	360	360	500					
Year 4				-173	192	192	192	360	360	360	500				
Year 5					-173	192	192	192	360	360	360	500			
Year 6						-173	192	192	192	360	360	360	500		
Year 7							-173	192	192	192	360	360	360	500	
Year 8								-173	192	192	192	360	360	360	500
Year 9									-173	192	192	192	360	360	360
Year 10										-173	192	192	192	360	360
Year 11											-173	192	192	192	360
Year 12												-173	192	192	192
Year 13													-173	192	192
Year 14														-173	192
Year 15															-173

Note: Estimated area of citrus plantings from 2012-2021 sourced from SunRISE Mapping and Research (June 2025). The NSW crop areas were derived from the Lower Murray-Darling NSW estimate presented in report, minus the crop areas recorded for the Lower Darling Menindee to Wentworth weir pool (River Reach 13 A) and Lower Murrumbidgee

Table 36 Estimated water usage of existing citrus plantings, not yet fully mature – New South Wales (ML)

Age of trees	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32
Year 1	-2,080	767	767	767	1,440	1,440	1,440	2,000							
Year 2		-2,080	1,054	1,054	1,054	1,980	1,980	1,980	2,750						
Year 3			-2,080	1,725	1,725	1,725	3,240	3,240	3,240	4,500					
Year 4				-2,080	1,917	1,917	1,917	3,600	3,600	3,600	5,000				
Year 5					-2,080	2,108	2,108	2,108	3,960	3,960	3,960	5,500			
Year 6						-2,080	2,204	2,204	2,204	4,140	4,140	4,140	5,750		
Year 7							-2,080	2,300	2,300	2,300	4,320	4,320	4,320	6,000	
Year 8								-2,080	2,300	2,300	2,300	4,320	4,320	4,320	6,000
Year 9									-2,080	2,300	2,300	2,300	4,320	4,320	4,320
Year 10										-2,080	2,300	2,300	2,300	4,320	4,320
Year 11											-2,080	2,300	2,300	2,300	4,320
Year 12												-2,080	2,300	2,300	2,300
Year 13													-2,080	2,300	2,300
Year 14														-2,080	2,300
Year 15															-2,080
Estimate of current water usage (2024-25)								**15,353**							
Estimate of water usage at maturity (2031-32)															**23,780**
Estimate of additional water usage of existing plantings															**8,428**

Note: Water application rates for citrus trees from year 1 to year 7 were sourced from the discussions with industry professionals (see above).

Estimated additional water usage of existing grape plantings per year – New South Wales, lower Murray

Table 37 Estimated area of grape plantings 2017-18 to 2024-25 – New South Wales (ha)

Age of trees	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32
Year 1					48	48	48	0							
Year 2						48	48	48	0						
Year 3							48	48	48	0					
Year 4								48	48	48	0				
Year 5									48	48	48	0			
Year 6										48	48	48	0		
Year 7											48	48	48	0	
Year 8												48	48	48	0
Year 9													48	48	48
Year 10														48	48
Year 11															48
Year 12															
Year 13															
Year 14															
Year 15															

Note: Estimated area of table grape plantings from 2012-2024 sourced from SunRISE Mapping and Research (June 2025). The NSW crop areas were derived from the Lower Murray-Darling NSW estimate presented in report, minus the crop areas recorded for the Lower Darling Menindee to Wentworth weir pool (River Reach 13 A) and Lower Murrumbidgee. Crop expansion estimates date back to 2021-22, in line with a 3-year maturation profile for table grapes. The Grapes (table, wine and dried) category includes grapevine – dried, grapevine – table and grapevine – wine categories.

Table 38 Estimated water usage of existing grape plantings, not yet fully mature – New South Wales (ML)

Age of trees	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32
Year 1					145	145	145								
Year 2						290	290	290							
Year 3							483	483	483						
Year 4								483	483	483					
Year 5									483	483	483				
Year 6										483	483	483			
Year 7											483	483	483		
Year 8												483	483	483	
Year 9													483	483	483
Year 10														483	483
Year 11															483
Year 12															
Year 13															
Year 14															
Year 15															
Estimate of current water usage (2024-25)								**1257**							
Estimate of water usage at maturity (2031-32)															**1,450**
Estimate of additional water usage of existing plantings															**193**

Note: Water application rates for grapes from year 1 to year 3 were sourced from discussions with industry professionals (see above). Water application rates vary across dried, wine and table grapes. However, due to data limitations an average water application rate was applied to all existing grape crops and potential grape developments.

Summary of additional water demand from existing plantings

Table 39 Summary of additional water demand from existing plantings in New South Wales (GL)

Crop	Estimated area of plantings yet to reach maturity (ha)	Estimated additional water demand from existing plantings (GL)
Almond	4,542	9.1
Citrus	1,982	8.4
Grape (table, wine & dried)	145	0.2
Total	**6,668**	**17.7**

Data Tables – Additional water usage existing plantings in Victoria

Estimated additional water usage of existing almond plantings per year – Victoria, lower Murray

Table 40 Estimated area of almond plantings 2017-18 to 2024-25 – Victoria (ha)

Age of trees	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32
Year 1	1,292	643	643	643	-13	-13	-13	0							
Year 2		1,292	643	643	643	-13	-13	-13	0						
Year 3			1,292	643	643	643	-13	-13	-13	0					
Year 4				1,292	643	643	643	-13	-13	-13	0				
Year 5					1,292	643	643	643	-13	-13	-13	0			
Year 6						1,292	643	643	643	-13	-13	-13	0		
Year 7							1,292	643	643	643	-13	-13	-13	0	
Year 8								1,292	643	643	643	-13	-13	-13	0
Year 9									1,292	643	643	643	-13	-13	-13
Year 10										1,292	643	643	643	-13	-13
Year 11											1,292	643	643	643	-13
Year 12												1,292	643	643	643
Year 13													1,292	643	643
Year 14														1,292	643
Year 15															1,292

Note: Estimated area of almond plantings from 2012-2024 sourced from SunRISE Mapping and Research (June 2025).

Table 41 Estimated water usage of existing almond plantings, not yet fully mature – Victoria (ML)

Age of trees	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32
Year 1	3,229	1,608	1,608	1,608	-187	-187	-187								
Year 2		6,458	3,217	3,217	3,217	-187	-187	-187							
Year 3			12,271	6,112	6,112	6,112	-187	-187	-187						
Year 4				13,563	6,755	6,755	6,755	-187	-187	-187					
Year 5					14,854	7,398	7,398	7,398	-187	-187	-187				
Year 6						16,146	8,042	8,042	8,042	-187	-187	-187			
Year 7							16,792	8,363	8,363	8,363	-187	-187	-187		
Year 8								18,083	9,007	9,007	9,007	-187	-187	-187	
Year 9									18,083	9,007	9,007	9,007	-187	-187	-187
Year 10										18,083	9,007	9,007	9,007	-187	-187
Year 11											18,083	9,007	9,007	9,007	-187
Year 12												18,083	9,007	9,007	9,007
Year 13													18,083	9,007	9,007
Year 14														18,083	9,007
Year 15															18,083
Estimate of current water usage (2024-25)								**41,327**							
Estimate of water usage at maturity (2031-32)															**44,543**
Estimate of additional water usage of existing plantings															**3,217**

Note: Water application rates for almond trees from year 1 to year 8 were sourced from the Almond Board of Australia during the 2020 analysis (see above).).

Estimated additional water usage of existing citrus plantings per year – Victoria, lower Murray

Table 42 Estimated area of citrus plantings 2017-18 to 2024-25 – Victoria (ha)

Age of trees	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32
Year 1	123	188	188	188	310	310	310	0							
Year 2		123	188	188	188	310	310	310	0						
Year 3			123	188	188	188	310	310	310	0					
Year 4				123	188	188	188	310	310	310	0				
Year 5					123	188	188	188	310	310	310	0			
Year 6						123	188	188	188	310	310	310	0		
Year 7							123	188	188	188	310	310	310	0	
Year 8								123	188	188	188	310	310	310	0
Year 9									123	188	188	188	310	310	310
Year 10										123	188	188	188	310	310
Year 11											123	188	188	188	310
Year 12												123	188	188	188
Year 13													123	188	188
Year 14														123	188
Year 15															123

Note: Estimated area of citrus plantings from 2012-2024 sourced from SunRISE Mapping and Research (June 2025).

Table 43 Estimated water usage of existing citrus plantings, not yet fully mature – Victoria (ML)

Age of trees	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32
Year 1	493	753	753	753	1,240	1,240	1,240								
Year 2		678	1,036	1,036	1,036	1,705	1,705	1,705							
Year 3			1,110	1,695	1,695	1,695	2,790	2,790	2,790						
Year 4				1,233	1,883	1,883	1,883	3,100	3,100	3,100					
Year 5					1,357	2,072	2,072	2,072	3,410	3,410	3,410				
Year 6						1,418	2,166	2,166	2,166	3,565	3,565	3,565			
Year 7							1,480	2,260	2,260	2,260	3,720	3,720	3,720		
Year 8								1,480	2,260	2,260	2,260	3,720	3,720	3,720	
Year 9									1,480	2,260	2,260	2,260	3,720	3,720	3,720
Year 10										1,480	2,260	2,260	2,260	3,720	3,720
Year 11											1,480	2,260	2,260	2,260	3,720
Year 12												1,480	2,260	2,260	2,260
Year 13													1,480	2,260	2,260
Year 14														1,480	2,260
Year 15															1,480
Estimate of current water usage (2024-25)								**15,573**							
Estimate of water usage at maturity (2031-32)															**15,680**
Estimate of additional water usage of existing plantings															**107**

Note: Water application rates for citrus trees from year 1 to year 7 were sourced from the discussions with industry professionals (see above).

Estimated additional water usage of existing grape plantings per year – Victoria, lower Murray

Table 44 Estimated area of grape plantings 2017-18 to 2024-25 – Victoria (ha)

Age of trees	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32
Year 1					-198	-198	-198	-250							
Year 2						-198	-198	-198	-250						
Year 3							-198	-198	-198	-250					
Year 4								-198	-198	-198	-250				
Year 5									-198	-198	-198	-250			
Year 6										-198	-198	-198	-250		
Year 7											-198	-198	-198	-250	
Year 8												-198	-198	-198	-250
Year 9													-198	-198	-198
Year 10														-198	-198
Year 11															-198
Year 12															
Year 13															
Year 14															
Year 15															

Note: Estimated area of table grape plantings from 2012-2024 sourced from SunRISE Mapping and Research (June 2025). Crop expansion estimates date back to 2021-22, in line with a 3-year maturation profile for table grapes. The Grapes (table, wine and dried) category includes grapevine – dried, grapevine – table and grapevine – wine categories.

Table 45 Estimated water usage of existing grape plantings, not yet fully mature – Victoria (ML)

Age of trees	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32
Year 1					-1,983	-1,983	-1,983	-2,500							
Year 2						-1,983	-1,983	-1,983	-2,500						
Year 3							-1,983	-1,983	-1,983	-2,500					
Year 4								-1,983	-1,983	-1,983	-2,500				
Year 5									-1,983	-1,983	-1,983	-2,500			
Year 6										-1,983	-1,983	-1,983	-2,500		
Year 7											-1,983	-1,983	-1,983	-2,500	
Year 8												-1,983	-1,983	-1,983	-2,500
Year 9													-1,983	-1,983	-1,983
Year 10														-1,983	-1,983
Year 11															-1,983
Year 12															
Year 13															
Year 14															
Year 15															
Estimate of current water usage (2024-25)								**-8,450**							
Estimate of water usage at maturity (2031-32)															**-8,450**
Estimate of additional water usage of existing plantings															**0**

Note: Water application rates for grapes from year 1 to year 3 were sourced from discussions with industry professionals (see above). Water application rates vary across dried, wine and table grapes. However, due to data limitations an average water application rate was applied to all existing grape crops and potential grape developments.

Summary of additional water demand from existing plantings

Table 46 Summary of additional water demand from existing plantings in Victoria (GL)

Crop	Estimated area of plantings yet to reach maturity (ha)	Estimated additional water demand from existing plantings (GL)
Almond	3,182	3.2
Citrus	1,618	0.1
Grape (table, wine & dried)	-845	0.0
Total	**3,955**	**3.3**

Data Tables – Additional water usage existing plantings in South Australia

Estimated additional water usage of existing almond plantings per year – South Australia, lower Murray

Table 47 Estimated area of almond plantings 2017-18 to 2024-25 – South Australia (ha)

Age of trees	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32
Year 1	1,237	873	873	873	1,007	1,007	1,007	2,000							
Year 2		1,237	873	873	873	1,007	1,007	1,007	2,000						
Year 3			1,237	873	873	873	1,007	1,007	1,007	2,000					
Year 4				1,237	873	873	873	1,007	1,007	1,007	2,000				
Year 5					1,237	873	873	873	1,007	1,007	1,007	2,000			
Year 6						1,237	873	873	873	1,007	1,007	1,007	2,000		
Year 7							1,237	873	873	873	1,007	1,007	1,007	2,000	
Year 8								1,237	873	873	873	1,007	1,007	1,007	2,000
Year 9									1,237	873	873	873	1,007	1,007	1,007
Year 10										1,237	873	873	873	1,007	1,007
Year 11											1,237	873	873	873	1,007
Year 12												1,237	873	873	873
Year 13													1,237	873	873
Year 14														1,237	873
Year 15															1,237

Note: Estimated area of almond plantings from 2012-2024 sourced from SunRISE Mapping and Research (June 2025).

Table 48 Estimated water usage of existing almond plantings, not yet fully mature – South Australia (ML)

Age of trees	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32
Year 1	3,092	2,183	2,183	2,183	2,517	2,517	2,517	5,000							
Year 2		6,183	4,367	4,367	4,367	5,033	5,033	5,033	10,000						
Year 3			11,748	8,297	8,297	8,297	9,563	9,563	9,563	19,000					
Year 4				12,985	9,170	9,170	9,170	10,570	10,570	10,570	21,000				
Year 5					14,222	10,043	10,043	10,043	11,577	11,577	11,577	23,000			
Year 6						15,458	10,917	10,917	10,917	12,583	12,583	12,583	25,000		
Year 7							16,077	11,353	11,353	11,353	13,087	13,087	13,087	26,000	
Year 8								17,313	12,227	12,227	12,227	14,093	14,093	14,093	28,000
Year 9									17,313	12,227	12,227	12,227	14,093	14,093	14,093
Year 10										17,313	12,227	12,227	12,227	14,093	14,093
Year 11											17,313	12,227	12,227	12,227	14,093
Year 12												17,313	12,227	12,227	12,227
Year 13													17,313	12,227	12,227
Year 14														17,313	12,227
Year 15															17,313
Estimate of current water usage (2024-25)								**79,793**							
Estimate of water usage at maturity (2031-32)															**124,273**
Estimate of additional water usage of existing plantings															**44,480**

Note: Water application rates for almond trees from year 1 to year 8 were sourced from the Almond Board of Australia during the 2020 analysis (see above).).

Estimated additional water usage of existing citrus plantings per year – South Australia, lower Murray

Table 49 Estimated area of citrus plantings 2017-18 to 2024-25 – South Australia (ha)

Age of trees	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32
Year 1	50	53	53	53	95	95	95	0							
Year 2		50	53	53	53	95	95	95	0						
Year 3			50	53	53	53	95	95	95	0					
Year 4				50	53	53	53	95	95	95	0				
Year 5					50	53	53	53	95	95	95	0			
Year 6						50	53	53	53	95	95	95	0		
Year 7							50	53	53	53	95	95	95	0	
Year 8								50	53	53	53	95	95	95	0
Year 9									50	53	53	53	95	95	95
Year 10										50	53	53	53	95	95
Year 11											50	53	53	53	95
Year 12												50	53	53	53
Year 13													50	53	53
Year 14														50	53
Year 15															50

Note: Estimated area of citrus plantings from 2012-2024 sourced from SunRISE Mapping and Research (June 2025).

Table 50 Estimated water usage of existing citrus plantings, not yet fully mature – South Australia (ML)

Age of trees	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32
Year 1	200	213	213	213	380	380	380								
Year 2		275	293	293	293	523	523	523							
Year 3			450	480	480	480	855	855	855						
Year 4				500	533	533	533	950	950	950					
Year 5					550	587	587	587	1,045	1,045	1,045				
Year 6						575	613	613	613	1,093	1,093	1,093			
Year 7							600	640	640	640	1,140	1,140	1,140		
Year 8								600	640	640	640	1,140	1,140	1,140	
Year 9									600	640	640	640	1,140	1,140	1,140
Year 10										600	640	640	640	1,140	1,140
Year 11											600	640	640	640	1,140
Year 12												600	640	640	640
Year 13													600	640	640
Year 14														600	640
Year 15															600
Estimate of current water usage (2024-25)								**4,768**							
Estimate of water usage at maturity (2031-32)															**5,940**
Estimate of additional water usage of existing plantings															**1,173**

Note: Water application rates for citrus trees from year 1 to year 7 were sourced from the discussions with industry professionals (see above).

Estimated additional water usage of existing grape plantings per year – South Australia, lower Murray

Table 51 Estimated area of grape plantings 2017-18 to 2024-25 – South Australia (ha)

Age of trees	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32
Year 1					-645	-645	-645	-1,250							
Year 2						-645	-645	-645	-1,250						
Year 3							-645	-645	-645	-1,250					
Year 4								-645	-645	-645	-1,250				
Year 5									-645	-645	-645	-1,250			
Year 6										-645	-645	-645	-1,250		
Year 7											-645	-645	-645	-1,250	
Year 8												-645	-645	-645	-1,250
Year 9													-645	-645	-645
Year 10														-645	-645
Year 11															-645
Year 12															
Year 13															
Year 14															
Year 15															

Note: Estimated area of table grape plantings from 2012-2024 sourced from SunRISE Mapping and Research (June 2025). Crop expansion estimates date back to 2021-22, in line with a 3-year maturation profile for table grapes. The Grapes (table, wine and dried) category includes grapevine – dried, grapevine – table and grapevine – wine categories.

Table 52 Estimated water usage of existing grape plantings, not yet fully mature – South Australia (ML)

Age of trees	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32
Year 1					-6,450	-6,450	-6,450	-12,500							
Year 2						-6,450	-6,450	-6,450	-12,500						
Year 3							-6,450	-6,450	-6,450	-12,500					
Year 4								-6,450	-6,450	-6,450	-12,500				
Year 5									-6,450	-6,450	-6,450	-12,500			
Year 6										-6,450	-6,450	-6,450	-12,500		
Year 7											-6,450	-6,450	-6,450	-12,500	
Year 8												-6,450	-6,450	-6,450	-12,500
Year 9													-6,450	-6,450	-6,450
Year 10														-6,450	-6,450
Year 11															-6,450
Year 12															
Year 13															
Year 14															
Year 15															

Age of trees	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32
Estimate of current water usage (2024-25)								**-31,850**							
Estimate of water usage at maturity (2031-32)															**-31,850**
Estimate of additional water usage of existing plantings															**0**

Note: Water application rates for grapes from year 1 to year 3 were sourced from discussions with industry professionals (see above). Water application rates vary across dried, wine and table grapes. However, due to data limitations an average water application rate was applied to all existing grape crops and potential grape developments.

Summary of additional water demand from existing plantings

Table 53 Summary of additional water demand from existing plantings in South Australia (GL)

Crop	Estimated area of plantings yet to reach maturity (ha)	Estimated additional water demand from existing plantings (GL)
Almond	8,877	44.5
Citrus	495	1.2
Grape (table, wine & dried)	-3,185	0.0
Total	**6,187**	**45.7**

Data tables: Summary of additional water demand from existing plantings across all states

Table 54 Summary of additional water demand from existing plantings across all states (note negative values indicate removed plantings whose effect of reduced water use is already accounted for in 2024-25)

Crop	Estimate area of plantings yet to reach maturity (ha)				Estimated additional water demand from existing plantings (GL)			
	New South Wales	Victoria	South Australia	Total	New South Wales	Victoria	South Australia	Total
Almond	4,542	3,182	8,877	**16,600**	9.1	3.2	44.5	**56.8**
Citrus	1,982	1,618	495	**4,095**	8.4	0.1	1.2	**9.7**
Grapes (table, wine & dried)	145	-845	-3,185	**-3,885**	0.2	0.0	0.0	**0.2**
Total	**6,668**	**3,955**	**6,187**	**16,810**	**17.7**	**3.3**	**45.7**	**66.7**

Data tables: Additional water demand from planned plantings

There are a number of known permanent horticultural developments in the connected Murray at various stages of planning. To estimate the total area and water demand for these potential developments, WSP spoke with industry professionals who have detailed on-the-ground knowledge.

The raw data for these planned plantings cannot be reported for confidentiality reasons. A summary of the planned plantings used to estimate projected permanent horticultural demand is shown below.

Table 55 Planned plantings - Estimated planned plantings (ha), connected Murray

Area	Plantings since the last SunRISE data collection (February 2024)		Planned Plantings	
	Total potential area (ha)	Total potential water demand (ML)	Total potential area (ha)	Total potential water demand (ML)
New South Wales – below the Choke	500	6000	2,801	41,010
Victoria	-250	-2500	5,621	71,634
South Australia	750	15500	-2,561	-25,610
Total	**1000**	**19000**	**5,861**	**87,034**

Table 56 Planned plantings - Estimated planned plantings (ha), lower Murray

Area	2024-25		Planned Plantings	
	Total potential area (ha)	Total potential water demand (ML)	Total potential area (ha)	Total potential water demand (ML)
New South Wales – below the Choke	500	6000	2,801	41,010
Victoria – below the Choke	-250	-2500	2,621	29,634

Area	2024-25		Planned Plantings	
	Total potential area (ha)	Total potential water demand (ML)	Total potential area (ha)	Total potential water demand (ML)
South Australia	750	15500	-2,561	-25,610
Total	**1000**	**19000**	**2,861**	**45,034**

Summary of permanent horticulture demand estimates

Table 57 Summary of permanent horticulture water demand estimates (GL)

	Connected Murray (GL)	Lower Murray (GL)
2024-25 water demand estimate	1,239.4	1,132.3
2024-25 water demand estimate (full maturity)	1,306.1	1,198.9
Projected water demand estimate (full maturity)	1,393.1	1,244.0

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